sEKV Parameter Extraction for the Generic 180nm Process

Short-channel pMOS (Version 1)

02.10.2026

In this notebook we extract the sEKV parameters for a short-channel pMOS transistor from the data generated with the EKV 2.6 compact model for a generic 180nm bulk CMOS process.

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1 Introduction

In this notebook we will extract the sEKV parameters [1] [2] for a short-channel pMOS transistors from a generic 180nm bulk CMOS process. The physical parameters are given in Table 1.1, the global process parameters in Table 1.2.

Table 1.1: Physical parameters

Parameter	Value	Unit
T	300	K
U_T	25.875	mV

Table 1.2: Global process parameters

Parameter	Value	Unit
V_{DD}	1.8	V
C_{ox}	8.46	$\frac{fF}{\mu m^2}$
W_{min}	200	nm
L_{min}	180	nm

The easiest way to extract the sEKV parameters is to use the python tool developed by H.C. Han and available on GitLab [3]. The tool and the extraction procedure are described in [4]. In this notebook we will detail the extraction procedure and show how the parameters can be extracted manually.

We start by looking at the channel width and length corrections for the drain current and for the capacitances. Then we will extract the sEKV parameter using a direct extraction methodology with the velocity parameter $\lambda_c = 0$. We then will extract the additional parameter λ_c using a direct extraction methodology. Then we will extract all the sEKV parameters by optimization using nonlinear curve fitting.

We also will extract the output conductance due to channel-length modulation (CLM) and the related parameter.

The extracted parameters are then all saved in an Excel worksheet.

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2 Transistor geometry parameters

2.1 Effective length and width for current

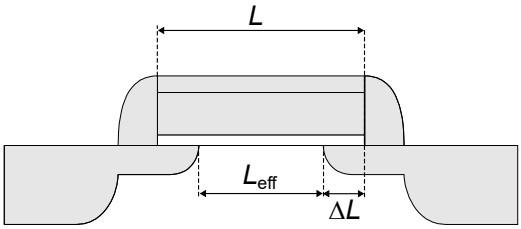


Figure 2.1: Definition of the effective length.

The effective length and width in the EKV 2.6 compact model are defined as [5]

$$W_{eff} = W + \Delta W, \tag{2.1}$$
$$L_{eff} = L + \Delta L, \tag{2.2}$$

where the width and length correction parameters ΔW and ΔL are given in Table 2.1.

Table 2.1: Transistor geometry.

Parameter	Symbol	Value	Unit
Selected transistor width and length			
Width	W	10	μm
Length	L	180	nm
Width and length correction parameters			
Width correction	ΔW	54	nm
Length correction	ΔL	-72	nm
Selected transistor effective width and length			
Effective width	W_{eff}	10.054	μm
Effective length	L_{eff}	108	nm

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3 DC Transfer Characteristic Parameters

3.1 Generating and importing the data

The data used for the sEKV parameters extraction is generated by simulation using the EKV 2.6 compact model [5] for a generic 180nm bulk CMOS process. We present the I_D-V_G and G_m-V_G data below.

3.1.1 I_D and G_m versus V_G

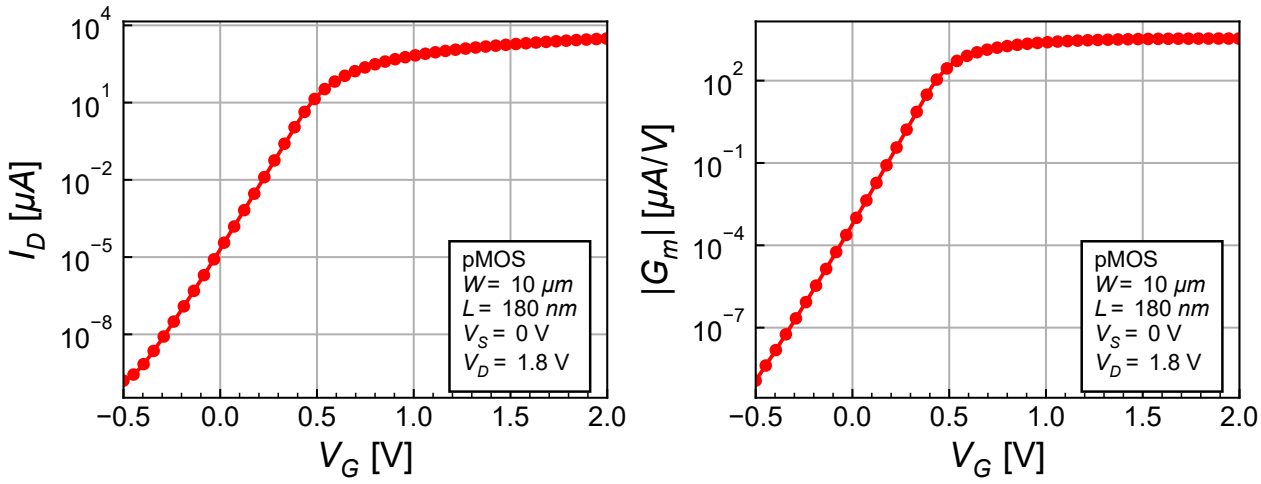


Figure 3.1: Imported I_D-V_G and G_m-V_G.

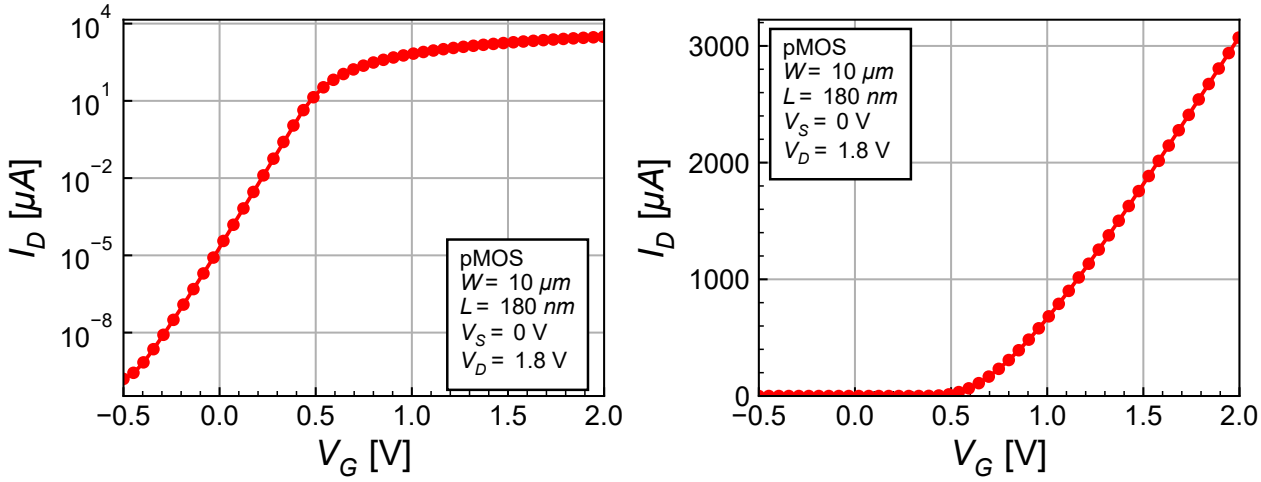


Figure 3.2: Imported I_D-V_G.

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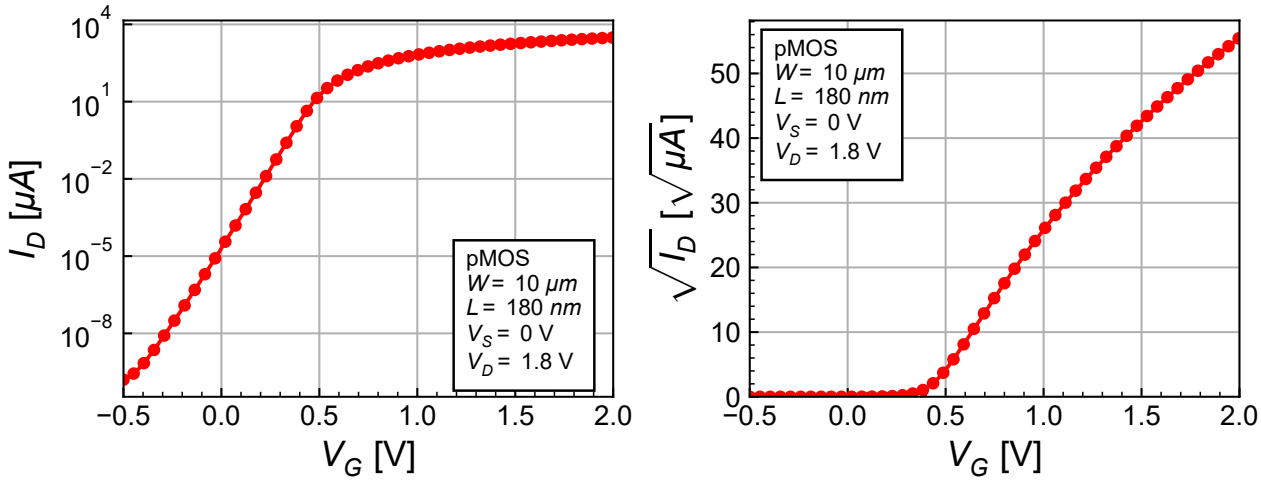


Figure 3.3: Imported I_D-V_G.

3.1.2 G_m-V_G and G_m-I_D

We now will check the derivative namely the gate transconductance obtained from the simulator and compare it to the numerical differentiation of the large-signal I_D-V_G characteristic.

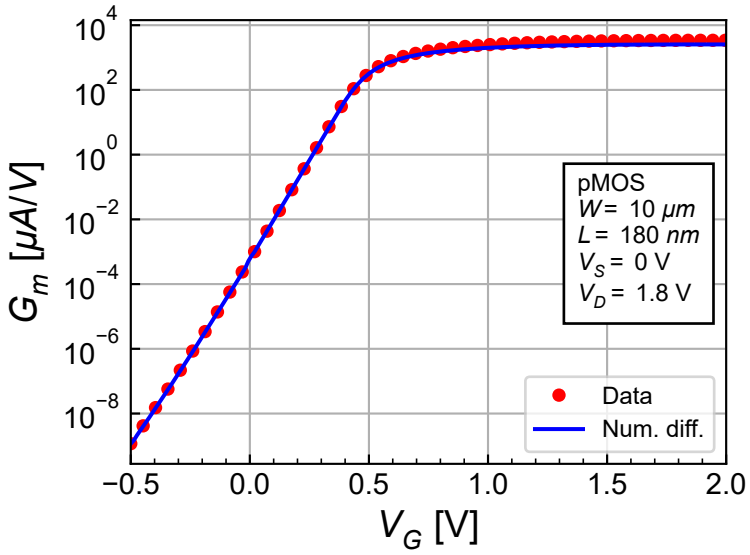


Figure 3.4: Check of the G_m-V_G consistency.

We see that the transconductance obtained by differentiating the large-signal I_D-V_G characteristic is equal to the transconductance extracted from the EKV 2.6 compact model. We will keep the value extracted from the EKV 2.6 compact model.

3.2 Filtering the outliers

Since the sEKV model [2] doesn't account for leakage currents such as gate induced drain leakage (GIDL) that appears at very low current, we need to filter the outlier points. This is done below with the outlier points shown in blue and the points used for the extraction shown in red.

We can now proceed with the parameter extraction, starting with the direct extraction approach with $\lambda_c = 0$.

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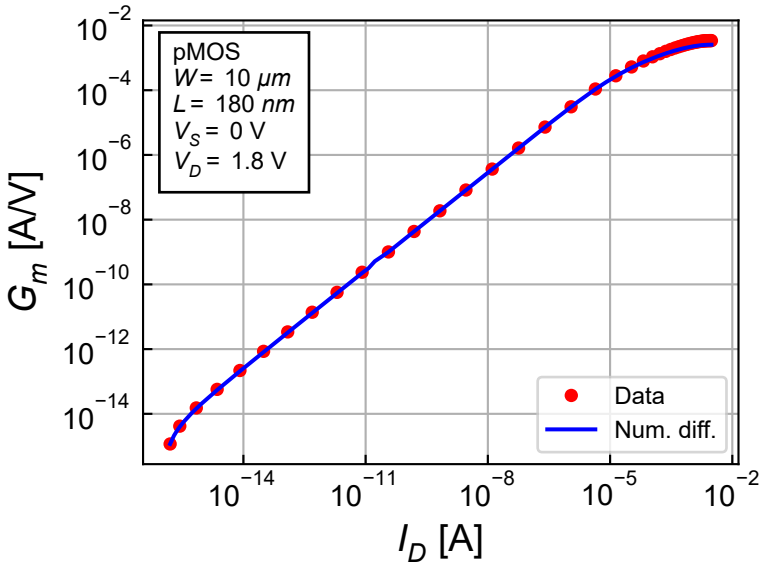


Figure 3.5: Check of the G_m-I_D consistency.

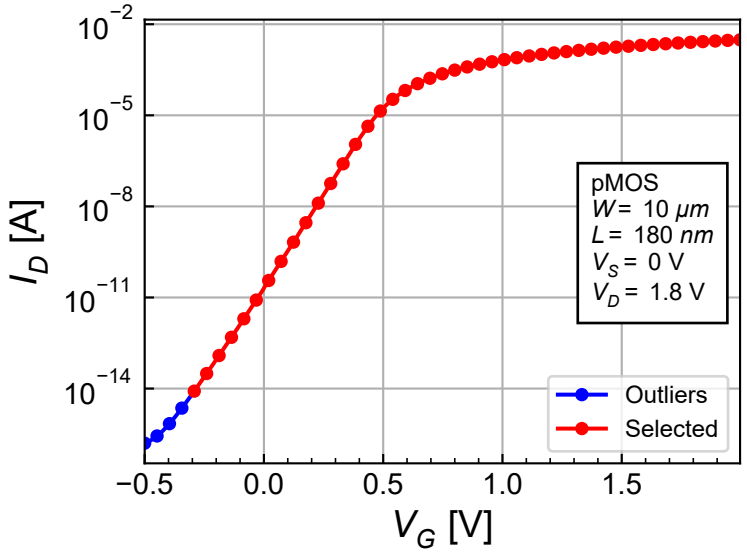


Figure 3.6: Filtering the outliers from the I_D-V_G charactersitic.

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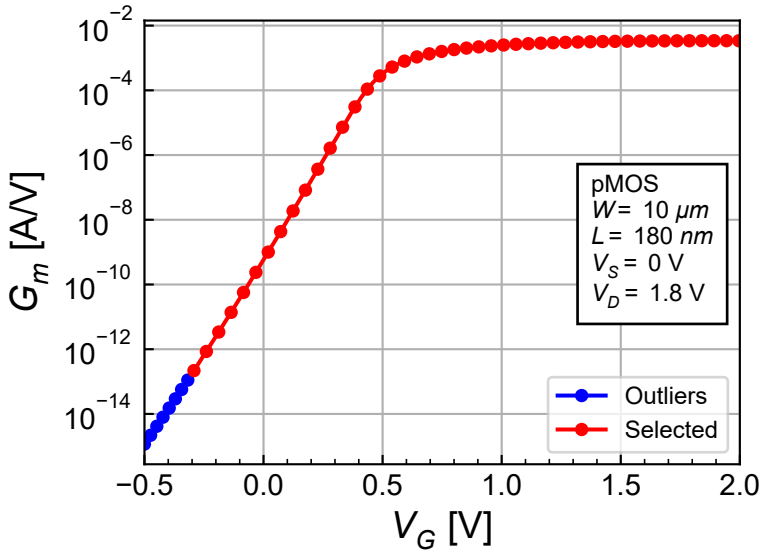


Figure 3.7: Filtering the outliers from the G_m-V_G charactersitic.

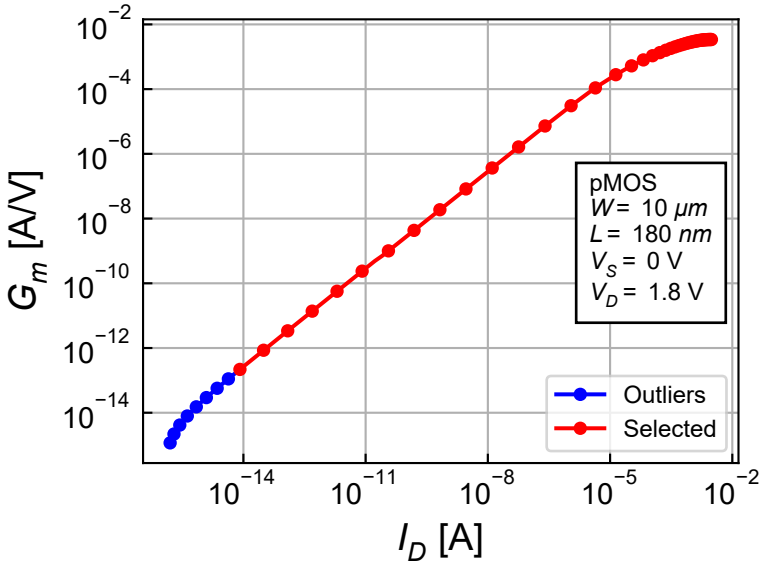


Figure 3.8: Filtering the outliers from the G_m-I_D charactersitic.

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3.3 Direct extraction with $\lambda_c = 0$

In this section we will extract the sEKV parametwers with $\lambda_c = 0$ and check the result on a short-channel transistor. In the direct extraction approach, we avoid using curve fitting or optimization and manipulate the data to extract a given parameter in a certain data range. We start extracting the slope factor n and the specific current I_{spec}.

3.3.1 Slope factor n and I_{spec} extraction

Remember that the gate transconductance in weak inversion and saturation is given by [1]

$$G_m = \frac{I_D}{n\,U_T}. \tag{3.1}$$

So if we plot $I_D/(G_m\,U_T)$ we should see a plateau in weak inversion the value of which is equal to the slope factor n as illustrated in Figure 3.9.

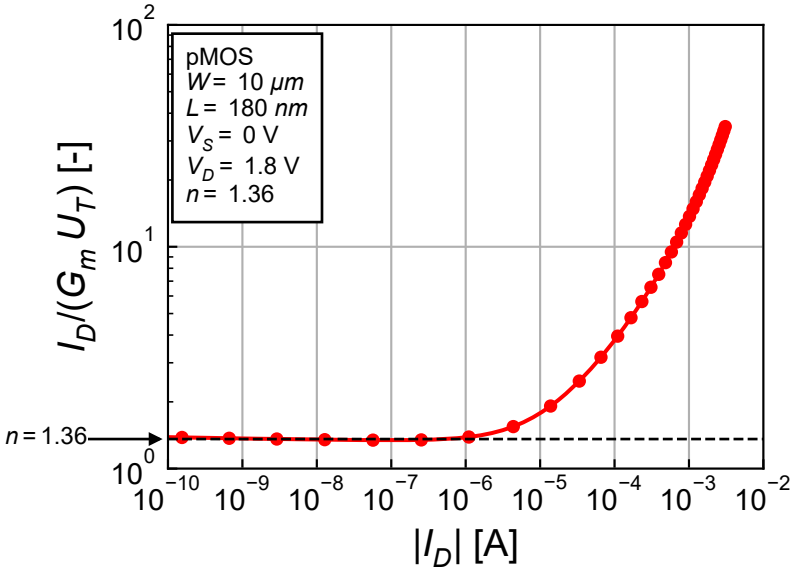


Figure 3.9: Slope factor extraction.

The plateau in Figure 3.9 corresponds to a slope factor $n = 1.36$. On the other hand the normalized G_m/I_D function for a long-channel transistor in strong inversion and saturation is given by [1]

$$\frac{G_m\,n\,U_T}{I_D} = \frac{1}{\sqrt{IC}} = \sqrt{\frac{I_{spec}}{I_D}}. \tag{3.2}$$

We can then plot $(G_m\,n\,U_T)^2/I_D$ which should find a maximum value equal to I_{spec}.

This is illustrated in Figure 3.10 resulting in $I_{spec} = 13.437\ \mu A$ corresponding to $I_{spec\square} = 144\ nA$. We can now plot $I_D/(G_m\,U_T)$ versus I_D as shown in Figure 3.11. We clearly see the two asymptotes in weak (i.e. $I_D < I_{spec}$) and strong inversion (i.e. $I_D > I_{spec}$).

Having extracted n and I_{spec}, we can now plot the normalized G_m/I_D function versus IC which is shown in Figure 3.12.

The fit is not good, particularly in strong inversion. This is expected for this minimum length transistor with $\lambda_c = 0$. We can slightly improve the fit in moderate inversion by shifting the curve to the right by increasing $I_{spec\square}$ as shown in Figure 3.13.

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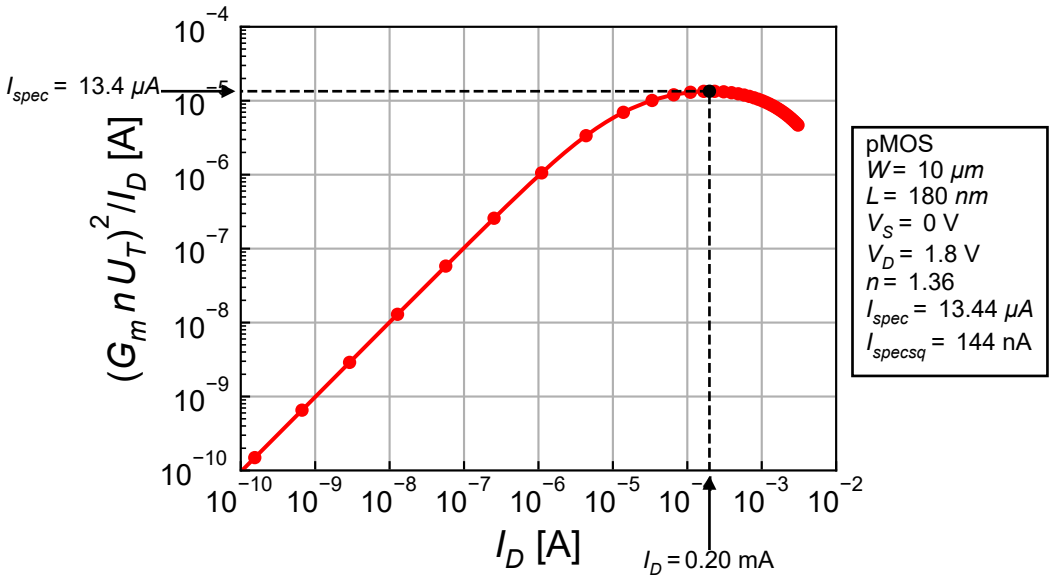


Figure 3.10: I_{spec} extraction.

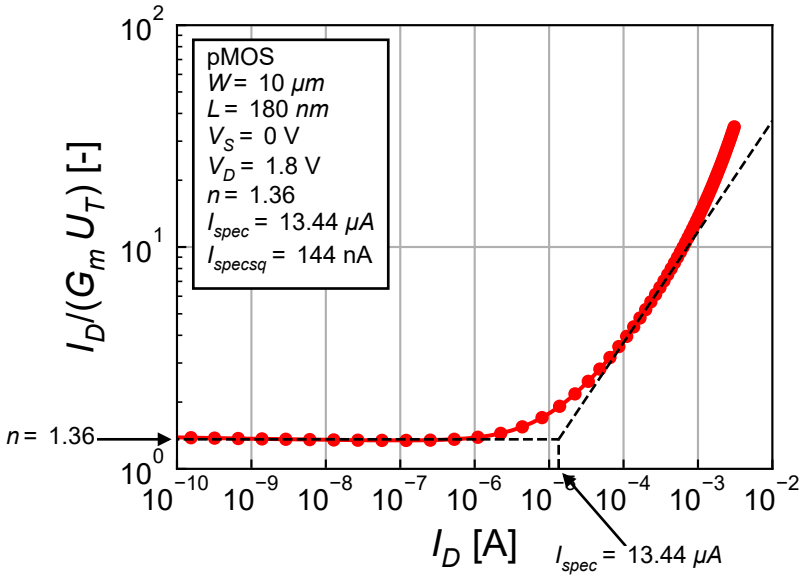


Figure 3.11: n and I_{spec} check.

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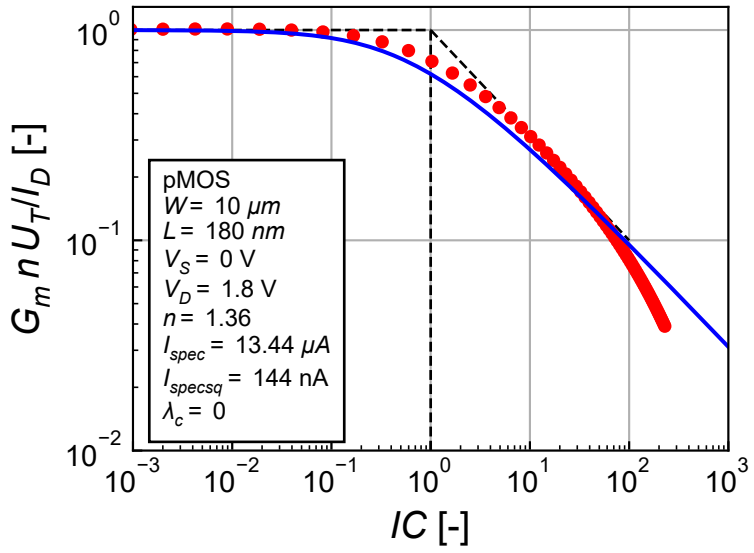


Figure 3.12: n and I_{spec} check.

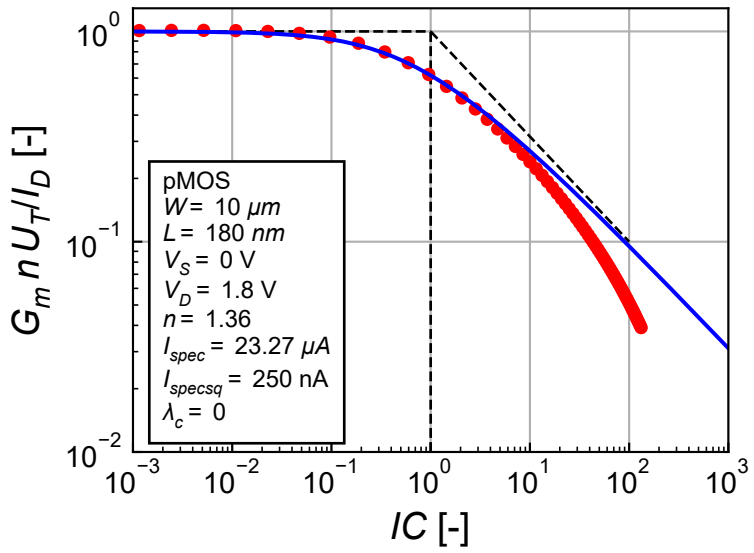


Figure 3.13: Fine tuning of the normalized G_m/I_D function versus IC in moderate inversion.

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The fit is now much better in moderate inversion but very bad in strong inversion. This is due to velocity saturation (VS) which is not accounted for when $\lambda_c = 0$. This simply demonstrates that for short-channel transistors we need to include the VS parameter λ_c. We will keep the updated value of $I_{spec\square}$ in the following extraction of the threshold voltage.

3.3.2 Threshold voltage extraction

We can extract the threshold voltage in weak inversion (assuming $V_S = 0$) from the normalized current (inversion coefficient) given by [1]

$$IC = e^{\frac{V_G - V_{T0}}{nU_T}}. \tag{3.3}$$

We can now plot

$$V_{T0} = V_G - nU_T \ln(IC) \tag{3.4}$$

to extract the threshold voltage. This results in the plot shown in Figure 3.14.

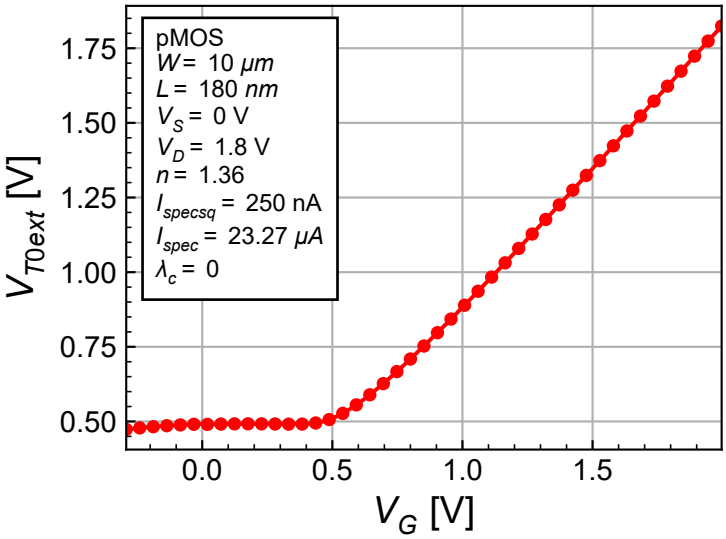


Figure 3.14: Threshold voltage extraction.

We see a plateau in weak inversion where we can average its value to get the threshold voltage in weak inversion which we can zoom into as shown in Figure 3.15.

The average value of V_{T0} in this range is given by $V_{T0} = 491\ mV$. We can now plot the I_D-V_G for the extracted parameters which is shown in Figure 3.16.

We get a good fit in weak and moderate inversion, but a strong deviation in strong inversion, which is expected since we focused on the moderate inversion and kept $\lambda_c = 0$.

Note that we can also extract the threshold voltage in strong inversion by plotting $\sqrt{I_D}$ versus V_G as shown in Figure 3.17.

We can then fit the linear portion of the curve and extract the intersection point on the V_G axis as shown in Figure 3.18 which results in $V_{T0} = 405\ mV$. We can check the I_D-V_G characersitic with this extracted threshold voltage in Figure 3.19.

As shown in Figure 3.19, we get a less good fit in weak inversion. We therefore keep the value of the threshold voltage extracted in weak inversion, namely $V_{T0} = 491\ mV$.

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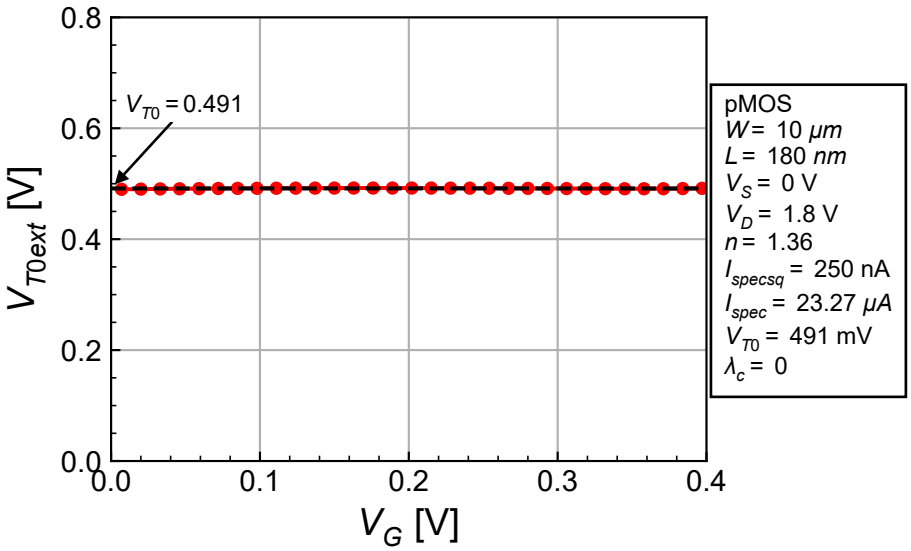


Figure 3.15: Threshold voltage extraction in weak inversion.

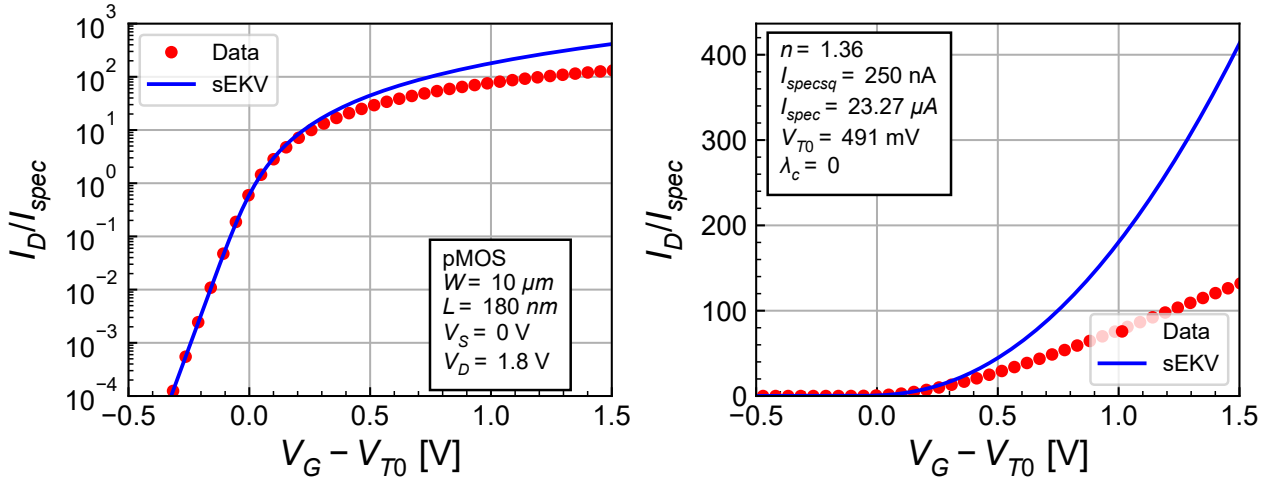


Figure 3.16: I_D-V_G for the extracted parameters.

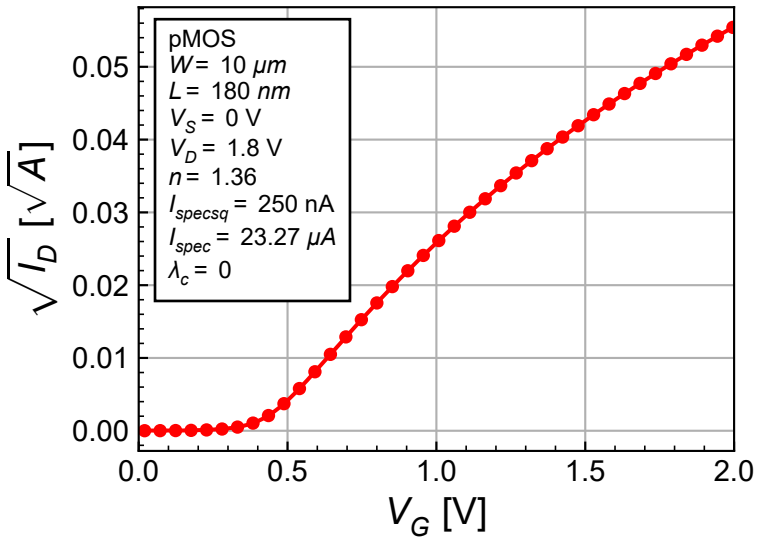


Figure 3.17: $\sqrt{I_D}$-V_G for the extracted parameters.

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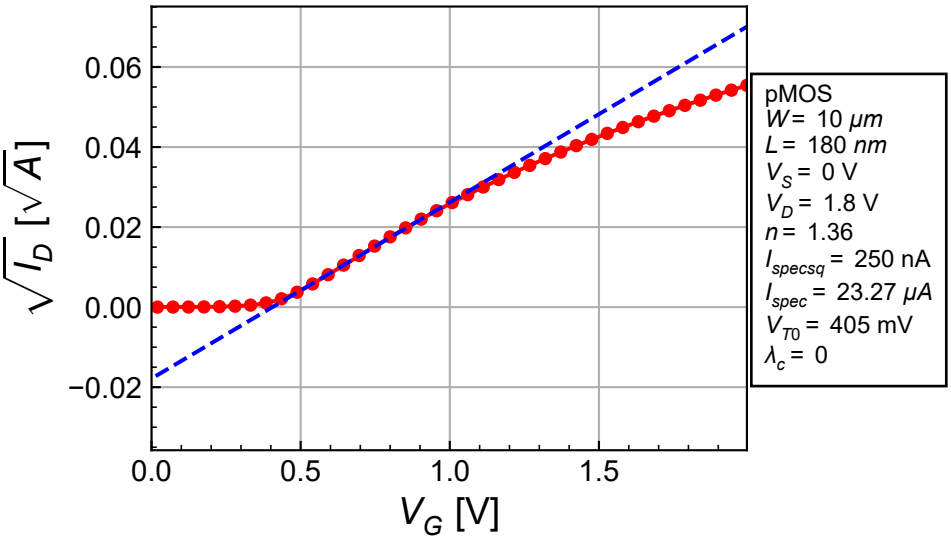


Figure 3.18: Threshold voltage extraction from $\sqrt{I_D}$-V_G in strong inversion.

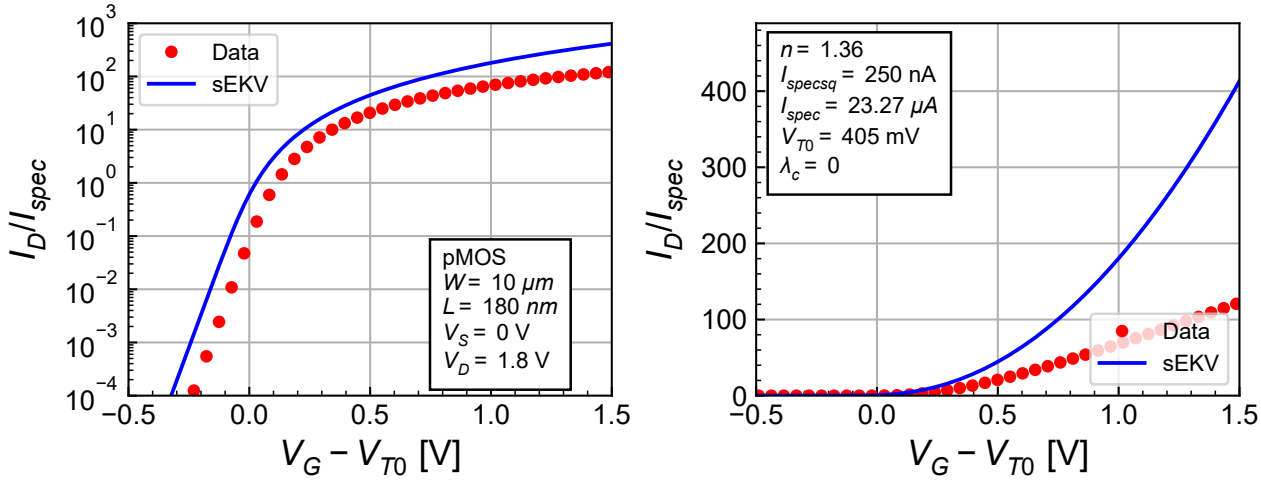


Figure 3.19: I_D-V_G for the extracted threshold voltage.

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3.3.3 Summary

The results of the direct extraction method with $\lambda_c = 0$ is shown in Figure 3.20, which includes the large-signal $IC = I_D/I_{spec}$ versus $V_G - V_{T0}$ on the left and the small-signal parameters $G_{ms}/G_{spec} = G_m\, n\, U_T/I_{spec}$ versus IC and $G_m\, n\, U_T/I_D$ versus IC on the right. We see an acceptable fit of the large- and small-signal characteristics in weak and moderate inversio. However the fir in very strong inversion (i.e. $100 < IC$) is very bad. The extracted parameters are summarized in Table 3.1.

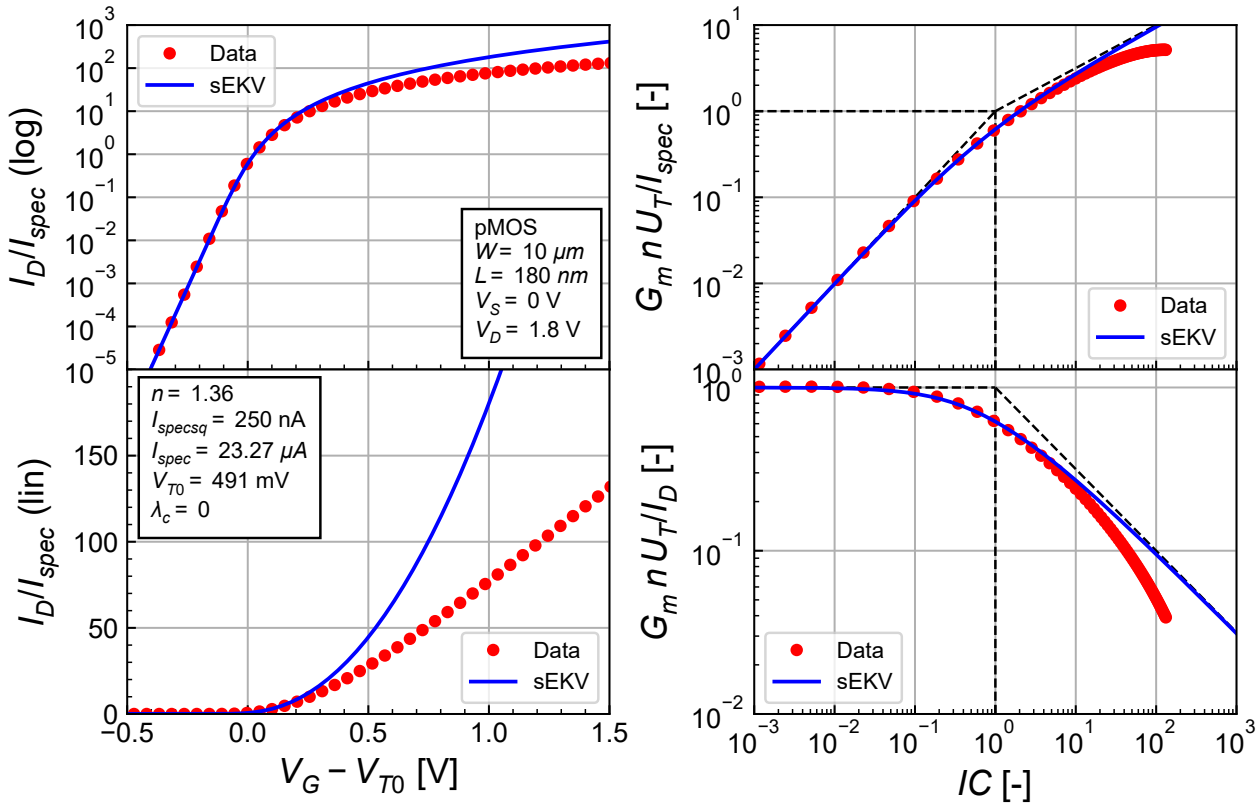


Figure 3.20: Summary of direct extraction with $\lambda_c = 0$.

Table 3.1: Direct extraction of the sEKV parameters with $\lambda_c = 0$.

Type	n	$I_{spec\square}$ $[nA]$	V_{T0} $[mV]$	λ_c	L_{sat} $[nm]$	Comment
pMOS	1.36	250	491	0	0	direct with $\lambda_c = 0$

This shows that fitting the large- and small-signal characteristic of a short transistor with $\lambda_c = 0$ does not give good results in strong inversion. We now will use the VS λ_c to get a better fit in strong inversion.

3.4 Direct extraction with $\lambda_c > 0$

3.4.1 Slope factor n extraction

Proceeding in the same way as above, we can extract the slope factor n from the characteristic shown in Figure 3.21.

We get the same value as before namely $n = 1.36$.

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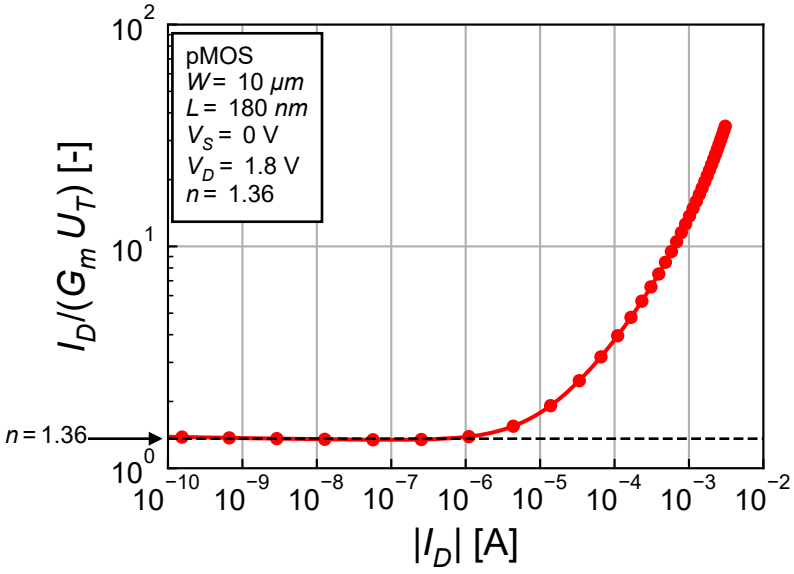


Figure 3.21: Slope factor extraction.

3.4.2 Specific current I_{spec} extraction

The specific current is extracted as above from the characteristic shown in Figure 3.22.

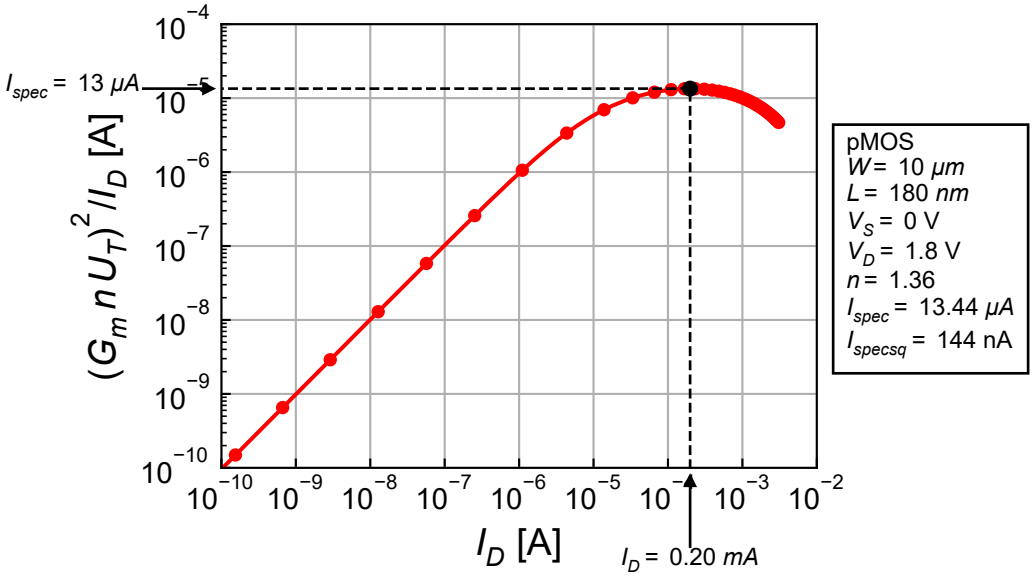


Figure 3.22: I_{spec} extraction.

We of course get the same value as before, namely $I_{spec} = 13.44\ \mu A$ corresponding to $I_{spec\square} = 144\ nA$. We can the plot $I_D/(G_m\, U_T)$ as shown in Figure 3.23. We see that the asymptote are correct except in strong inversion where the curve is steeper. This can then be handled by the additional VS parameter λ_c.

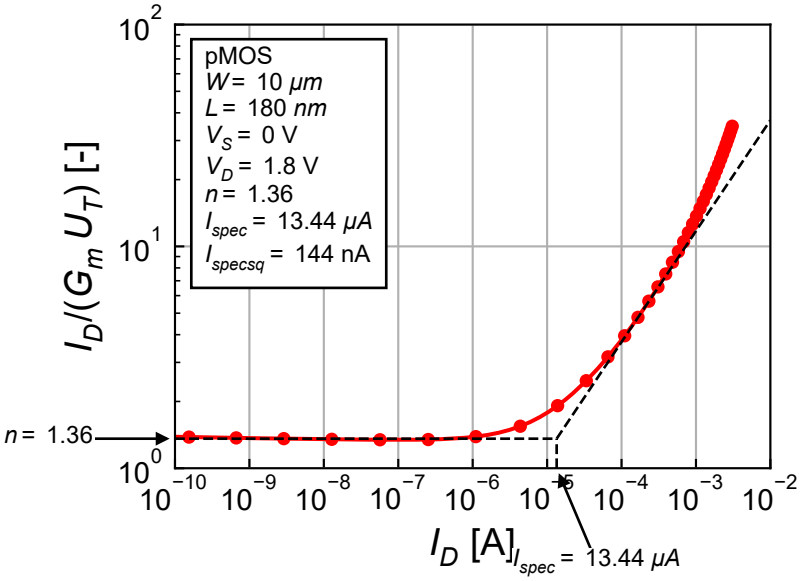


Figure 3.23: n and I_{spec} check.

3.4.3 Velocity saturation parameter λ_c extraction

We can extract λ_c by looking at the asymptote in very strong inversion. For a short-channel transistor in strong inversion and saturation, the normalized G_m/I_D is given by [1]

$$\frac{G_m\, n\, U_T}{I_D} = \frac{1}{\lambda_c\, IC} = \frac{I_{spec}}{\lambda_c\, I_D}. \tag{3.5}$$

So if we plot $I_{spec}/(G_m\, n\, U_T)$ it will have a minimum at λ_c as illustrated in Figure 3.24.

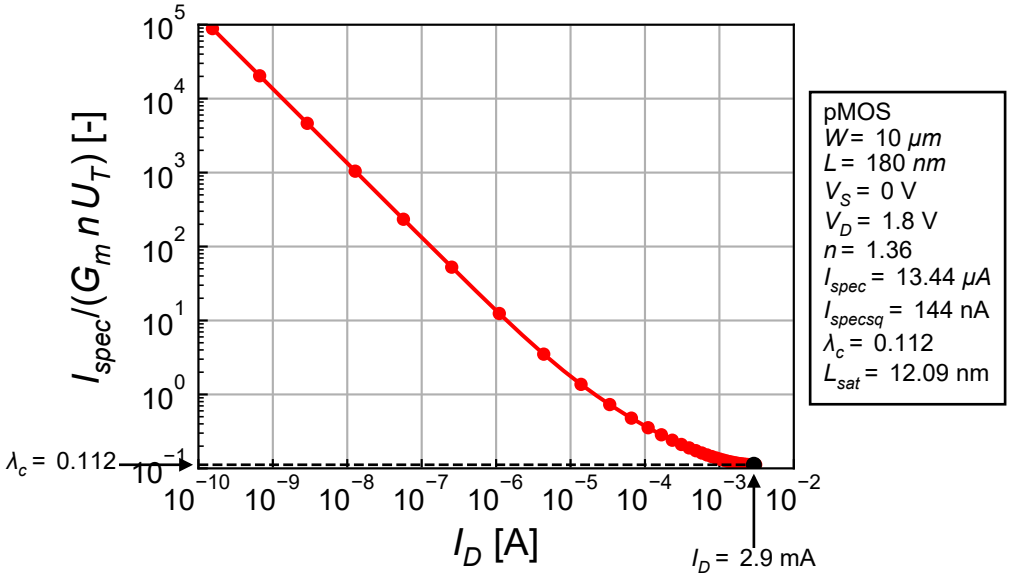


Figure 3.24: Extraction of λ_c.

This results in $\lambda_c = 0.112$ corresponding to $L_{sat} = 12.09\ nm$. We can now add the strong inversion asymptote on the $I_D/(G_m\, n\, U_T)$ versus I_D characteristic as shown in Figure 3.25.

We can now check the normalized G_m/I_D characteristic which is plotted in Figure 3.26.

The fit is OK at the asymptotes but not good in moderate inversion. We can try to improve the fit by shifting the curve to the right by increasing I_{spec} and also increase λ_c in order to bend the curve more

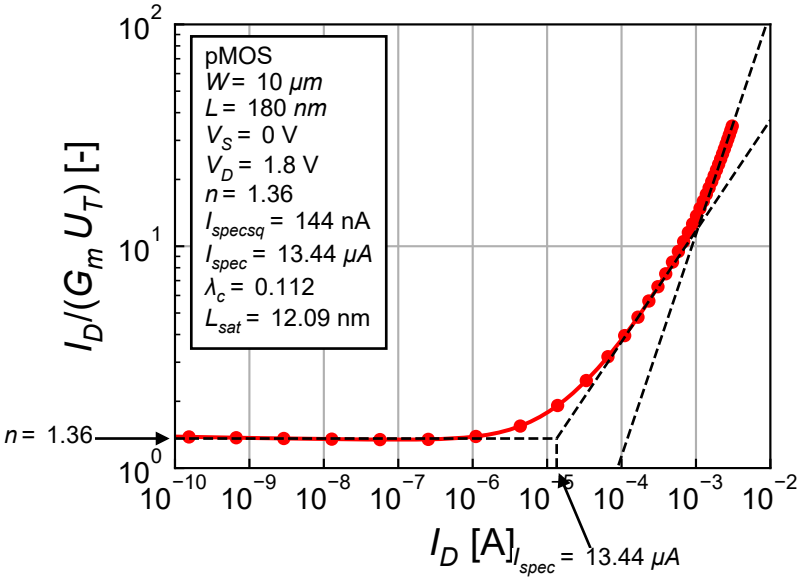


Figure 3.25: n, I_{spec} and λ_x check.

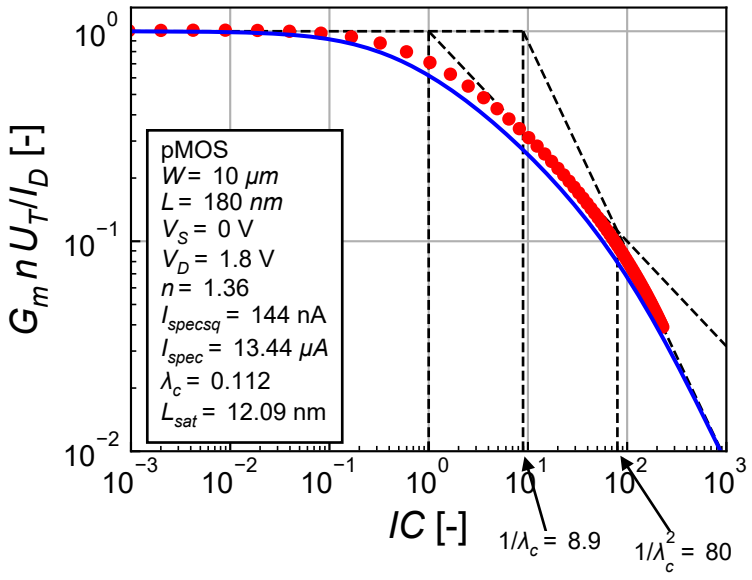


Figure 3.26: Check of the normalized G_m/I_D for n, I_{spec} and λ_c.

downwards in strong inversion and have a better fit in moderate and strong inversion as illustrated in Figure 3.27.

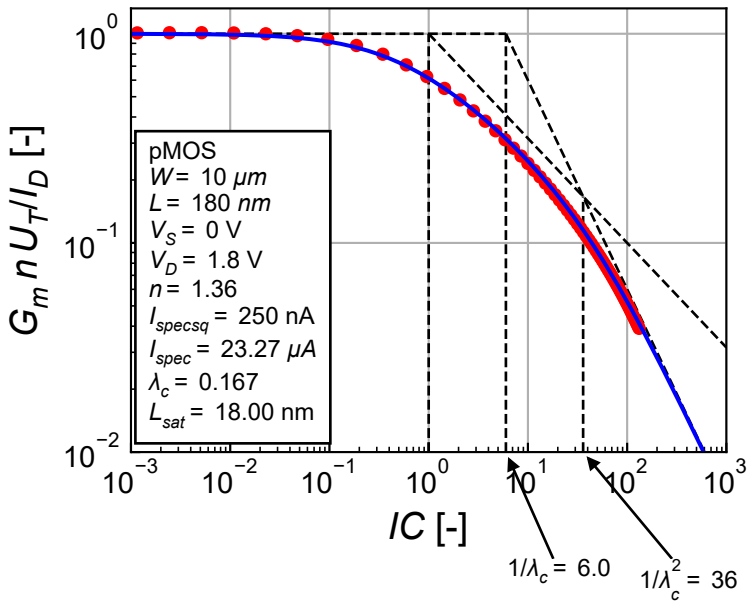


Figure 3.27: Tuning of I_{spec} and λ_c on the normalized G_m/I_D.

The fit is almost perfect in all regions of operations. However the parameters $I_{spec\square} = 250\ nA$ and $L_{sat} = 18\ nm$ are now fairly large and would probably not give a good fit of the long-channel transistor. We therefore reduce their values as shown in Figure 3.28.

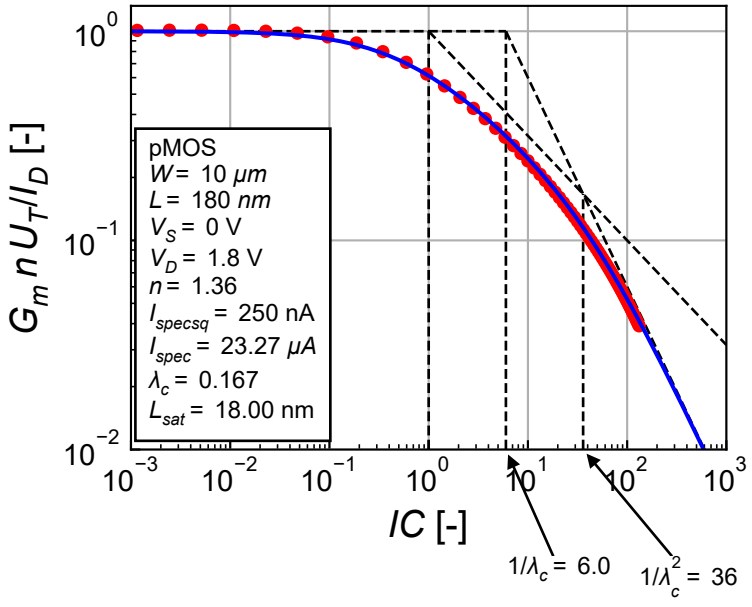


Figure 3.28: Tuning of I_{spec} and λ_c on the normalized G_m/I_D.

The fit is now OK, offering a good trade-off between moderate and strong inversion.

3.4.4 Threshold voltage extraction

We now extract the threshold voltage in the same way as above, first plotting V_{T0ext} versus V_G as shown in fig-vt0ext_vg.

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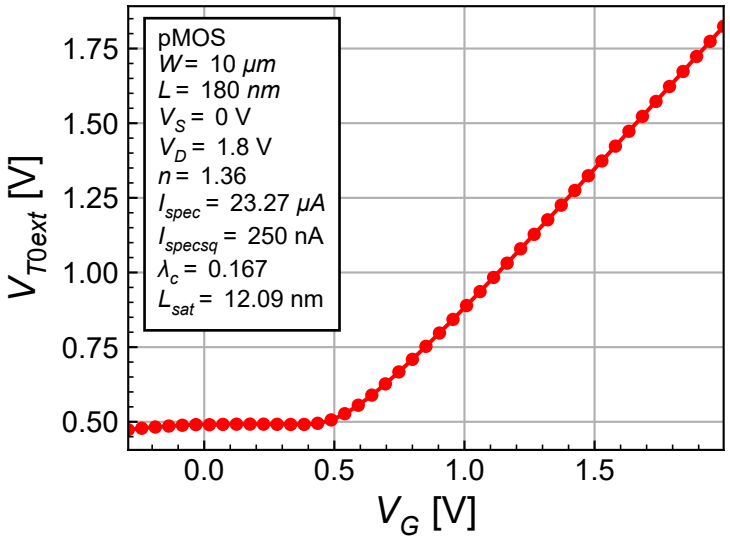


Figure 3.29: Threshold voltage extraction.

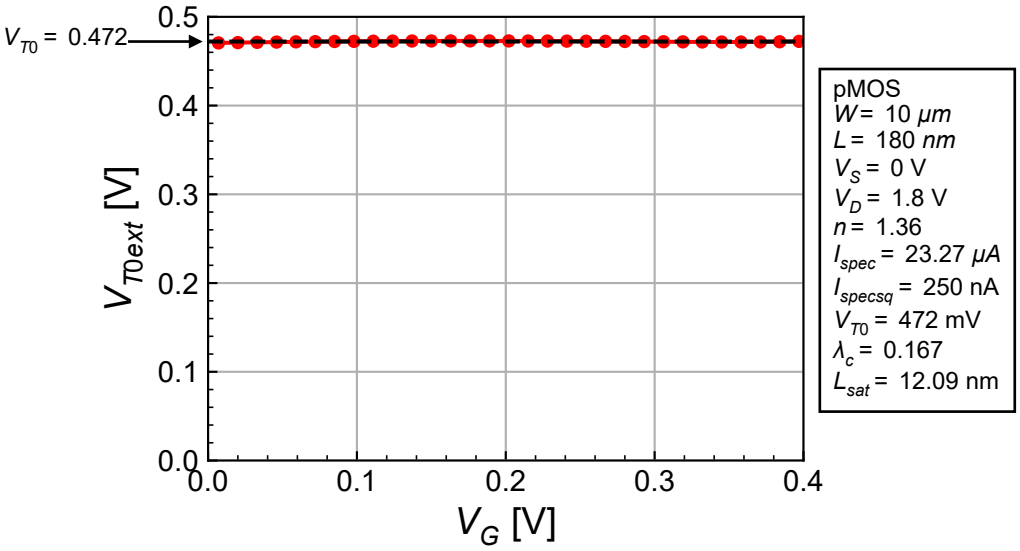


Figure 3.30: Threshold voltage extraction in weak inversion.

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Averaging the threshold voltage in the range where it is about constant results in $V_{T0} = 472\ mV$ which is slightly lower than the value extracted with $\lambda_c = 0$. We can now check the I_D-V_G curves which are shown in Figure 3.31.

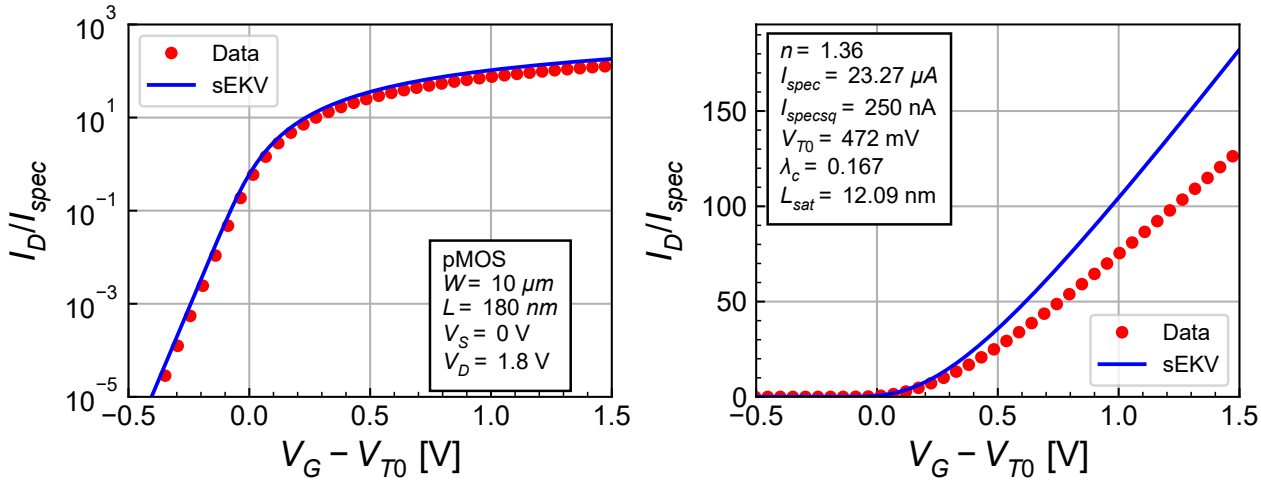


Figure 3.31: I_D-V_G for the extracted parameters.

We see that the threshold voltage is too low. We can fine tune it manually and adjust $I_{spec\square}$ and λ_c as well to get the fit shown in Figure 3.32.

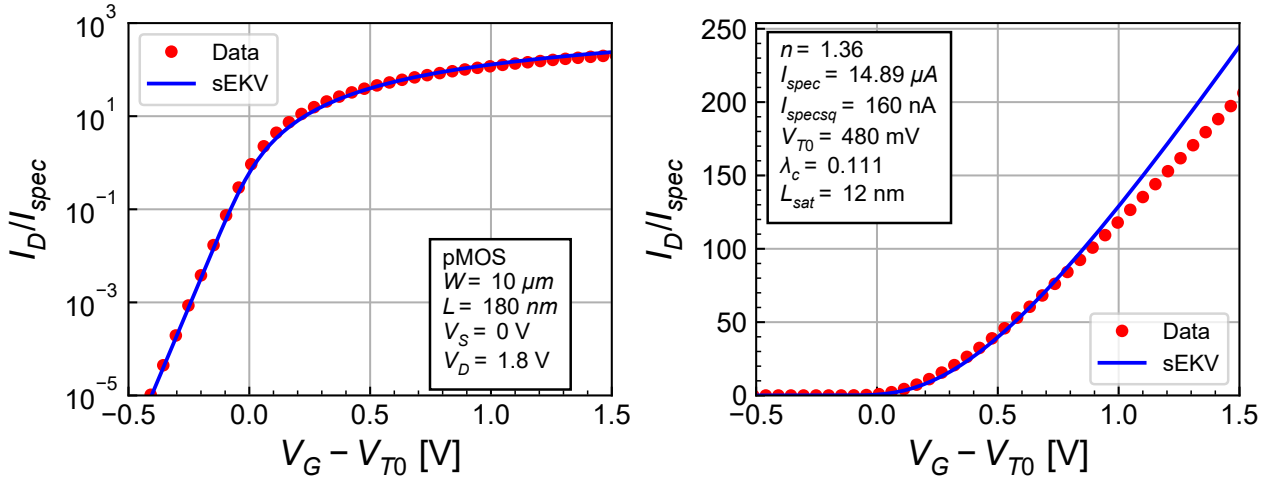


Figure 3.32: I_D-V_G for the extracted parameters.

We now have a very good fit of the I_D-V_G characteristic in the entire range of $V_G - V_{T0}$, from weak to strong inversion.

3.4.5 Summary

The results of the direct extraction with $\lambda_c > 0$ are summarized in Figure 3.33 for the sEKV parameters presented in Table 3.2.

We finally get a very good fit of all large- and small-signal characteristics with reasonably low values of $I_{spec\square} = 200\ nA$ and $L_{sat} = 15\ nm$ ($\lambda_c = 0.139$).

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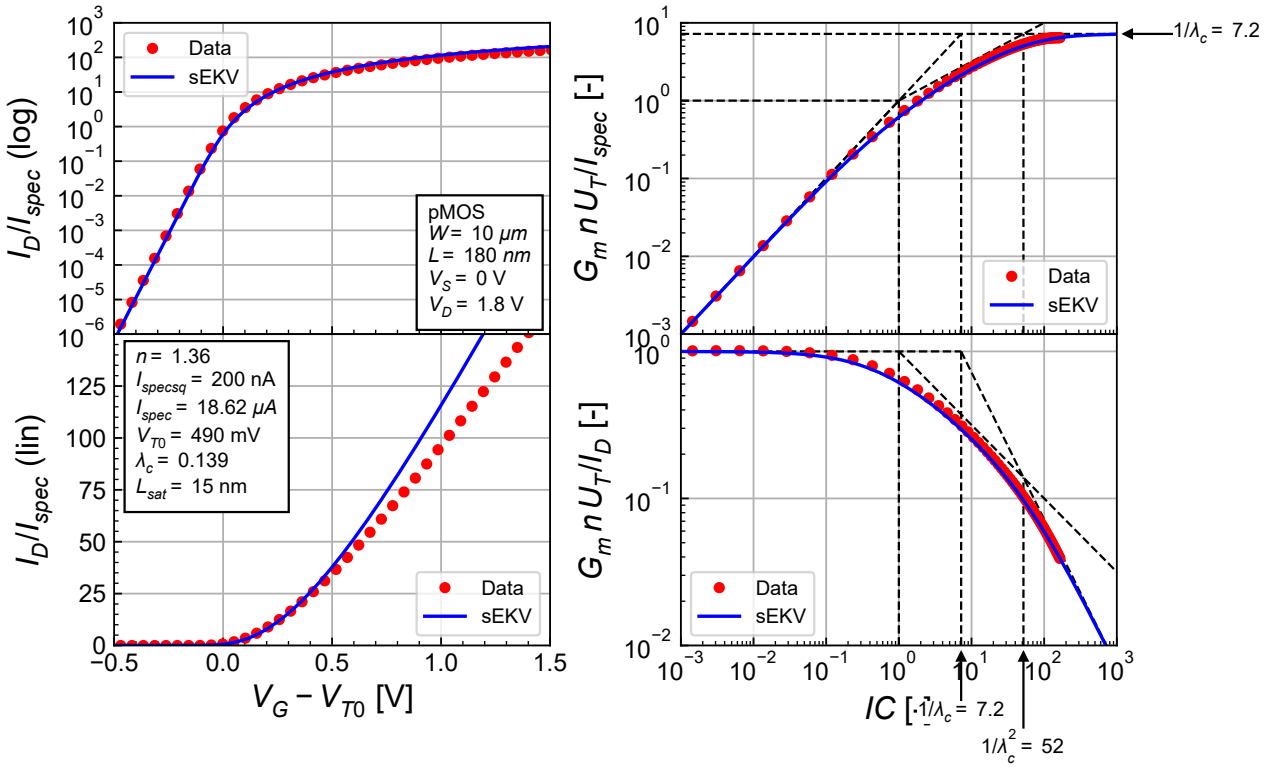


Figure 3.33: Summary of the direct extraction with $\lambda_c > 0$.

Table 3.2: Extraction of the sEKV parameters using direct extraction with $\lambda_c > 0$.

Type	n	$I_{spec\square}$ $[nA]$	V_{T0} $[mV]$	λ_c	L_{sat} $[nm]$	Comment
pMOS	1.36	250	491	0.00	0.00	direct with $\lambda_c = 0$
pMOS	1.36	200	490	0.14	15.00	direct with $\lambda_c > 0$

3.5 Extraction using curve fitting

3.5.1 Extraction of V_{T0}, I_{spec} and λ_c from the I_D-V_G characteristic

We can extract V_{T0}, I_{spec} and λ_c by curve fitting of the I_D-V_G characteristic and then check the result on the small-signal characteristic. For this we keep the value of the slope factor extracted above $n = 1.36$. The result is shown in Figure 3.34.

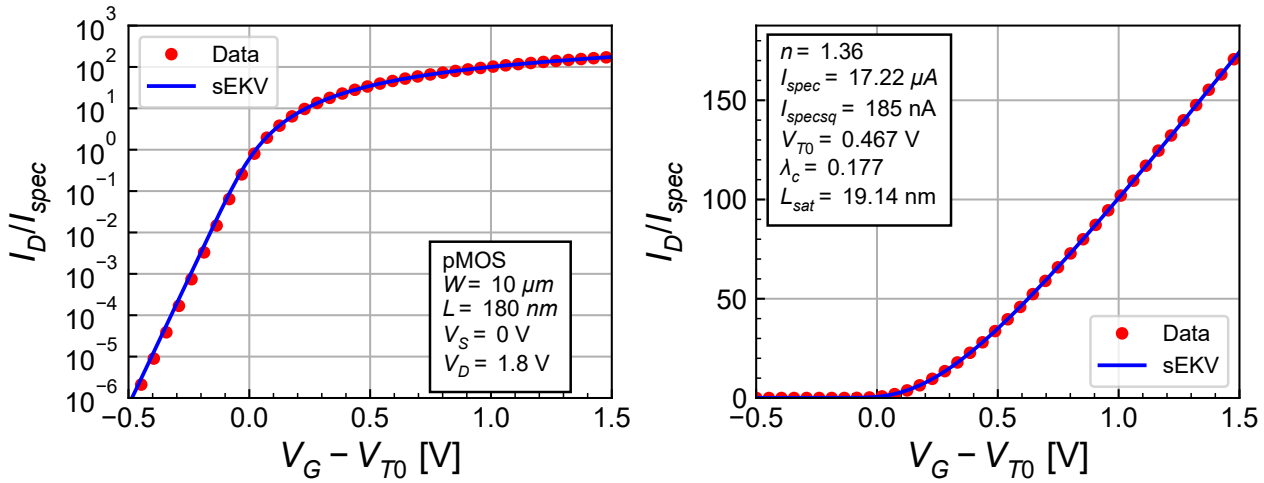


Figure 3.34: Extraction of V_{T0}, I_{spec} and λ_c from the I_D-V_G characteristic.

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The fit is very good with parameter values that are close to those extracted in the direct extraction

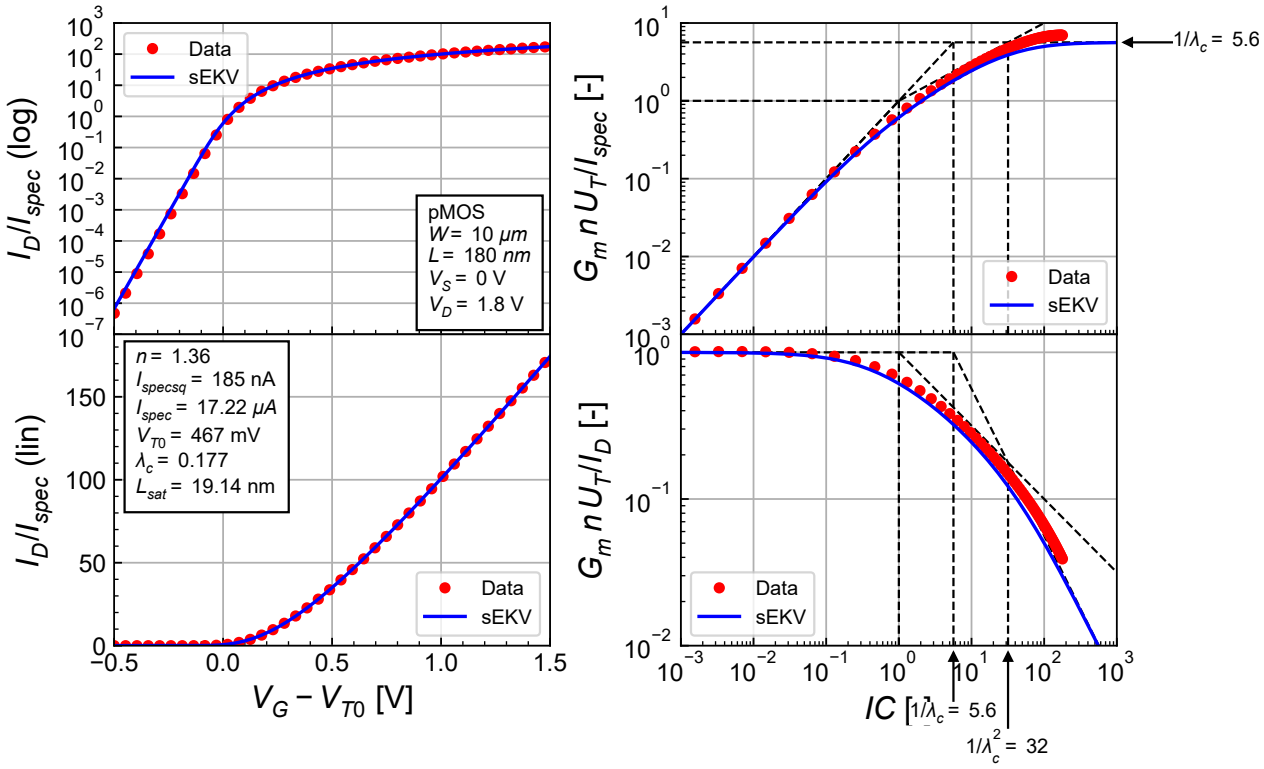


Figure 3.35: Summary of extraction by curve fitting.

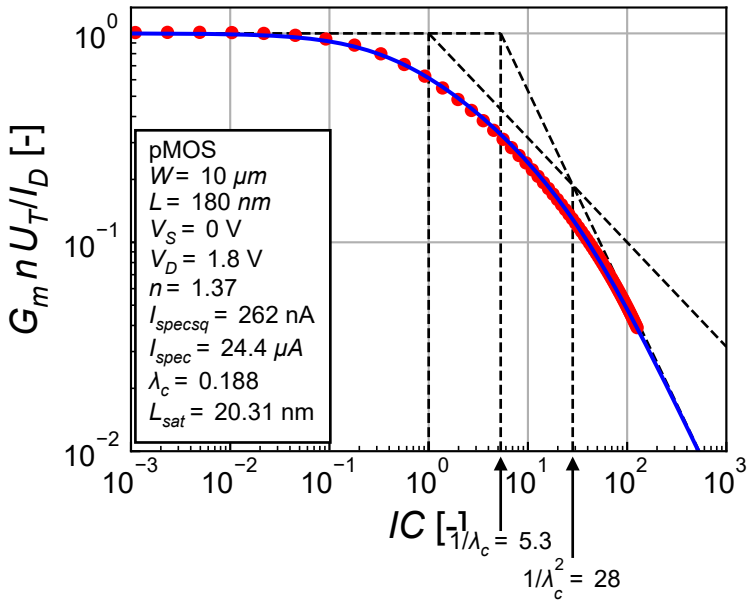


Figure 3.36: Extraction of n, I_{spec} and λ_c by curve fitting.

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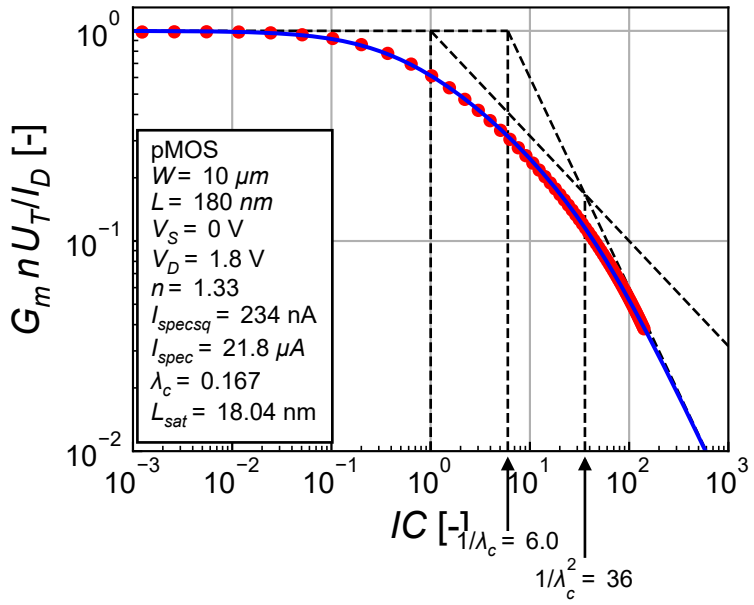


Figure 3.37: Extraction of I_{spec} and λ_c by curve fitting keeping n constant.

We can now extract the threshold voltage V_{T0}.

3.5.3 Threshold voltage extraction

We extract the threshold voltage from the I_D-V_G characteristic keeping the value of n, I_{spec} and λ_c extracted above. This result in the fit shown in Figure 3.38.

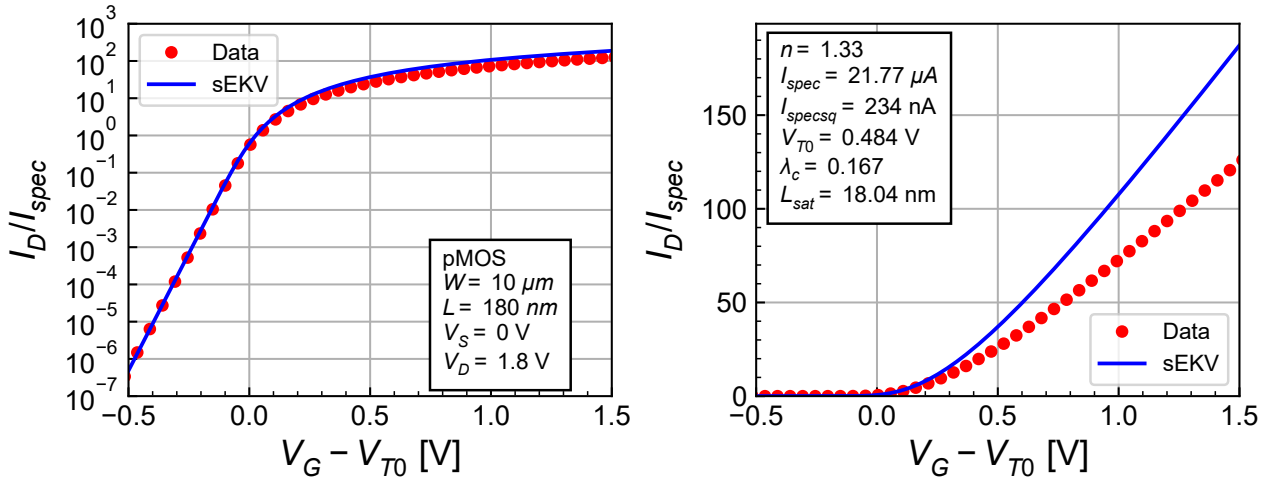


Figure 3.38: Threshold voltage extraction.

The I_D-V_G characteristic is off in strong inversion. We can correct this by extracting I_{spec} and λ_c on the I_D-V_G characteristic keeping V_{T0} and n. This results in the fit shown in Figure 3.39.

This results in $V_{T0} = 484\,mV$ which is close to what was obtained with the direct extraction. Figure 3.38 shows that curve fitting can result in a very good fit in all regions of operation!

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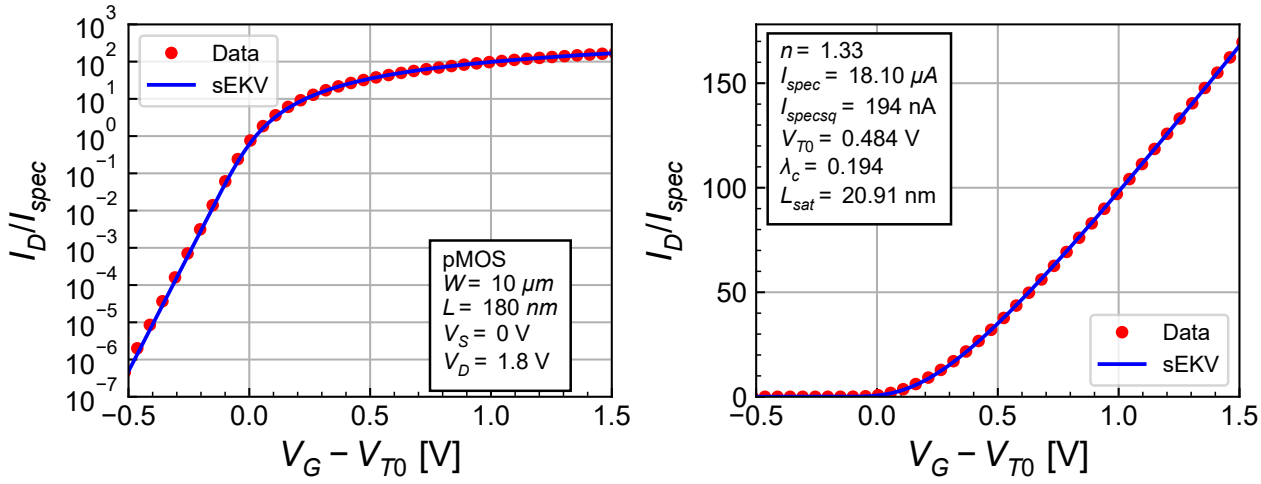


Figure 3.39: I_D-V_G.

3.5.4 Summary

The result of the curve fitting extraction methodology is summarized in Figure 3.40 with the paramters given in Table 3.3. We see that we obtained an overall good fit in all regions of operation.

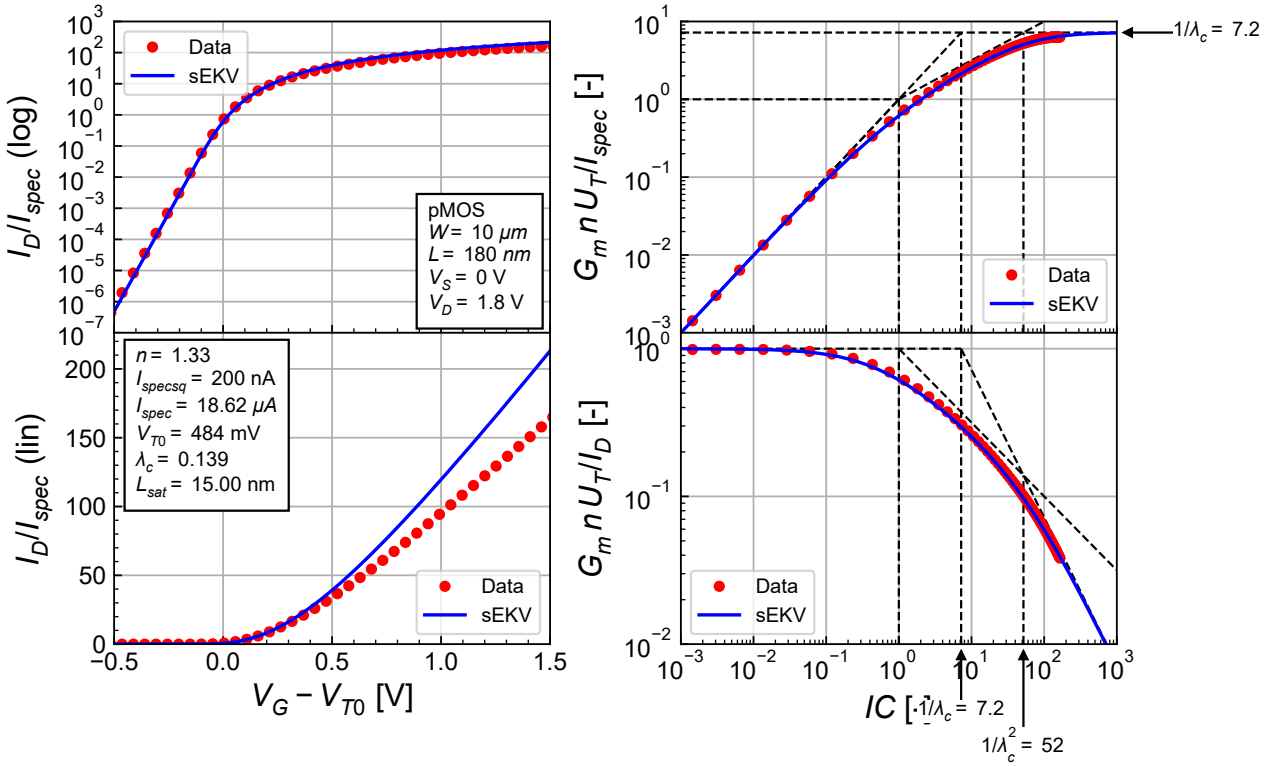


Figure 3.40: Summary of extraction by curve fitting for the long-channel.

Table 3.3: Extraction of the sEKV parameters using curve fitting with $\lambda_c > 0$.

Type	n	$I_{spec\square}$ $[nA]$	V_{T0} $[mV]$	λ_c	L_{sat} $[nm]$	Comment
pMOS	1.36	250	491	0.000	0.000	direct with $\lambda_c = 0$
pMOS	1.36	200	490	0.139	15.000	direct with $\lambda_c > 0$
pMOS	1.33	200	484	0.139	15.000	curve fitting with $\lambda_c > 0$

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3.6 Comparison with parameters extracted from the EKV 2.6 compact model

We can compare the data with the parameters taken directly from the EKV 2.6 compact model parameter deck. More precisely, V_{T0} is taken equal to the value used in EKV 2.6, namely $V_{T0} = 0.457$ V. The slope factor is calculated as

$$n = n_0 = 1 + \frac{\Gamma}{2\sqrt{\Psi_0}}, \tag{3.6}$$

where $\Gamma = 0.609\ \sqrt{V}$ and $\Psi_0 = 0.990\ V$. $I_{spec\square}$ is calculated as

$$I_{spec\square} = 2\ n\,\beta\,U_T^2, \tag{3.7}$$

with $\beta = 99\ \mu A/V^2$. The parameter L_{sat} is not included in EKV 2.6. We can set it to $\lambda_c = 0$ resulting in the fit shown in Figure 3.41.

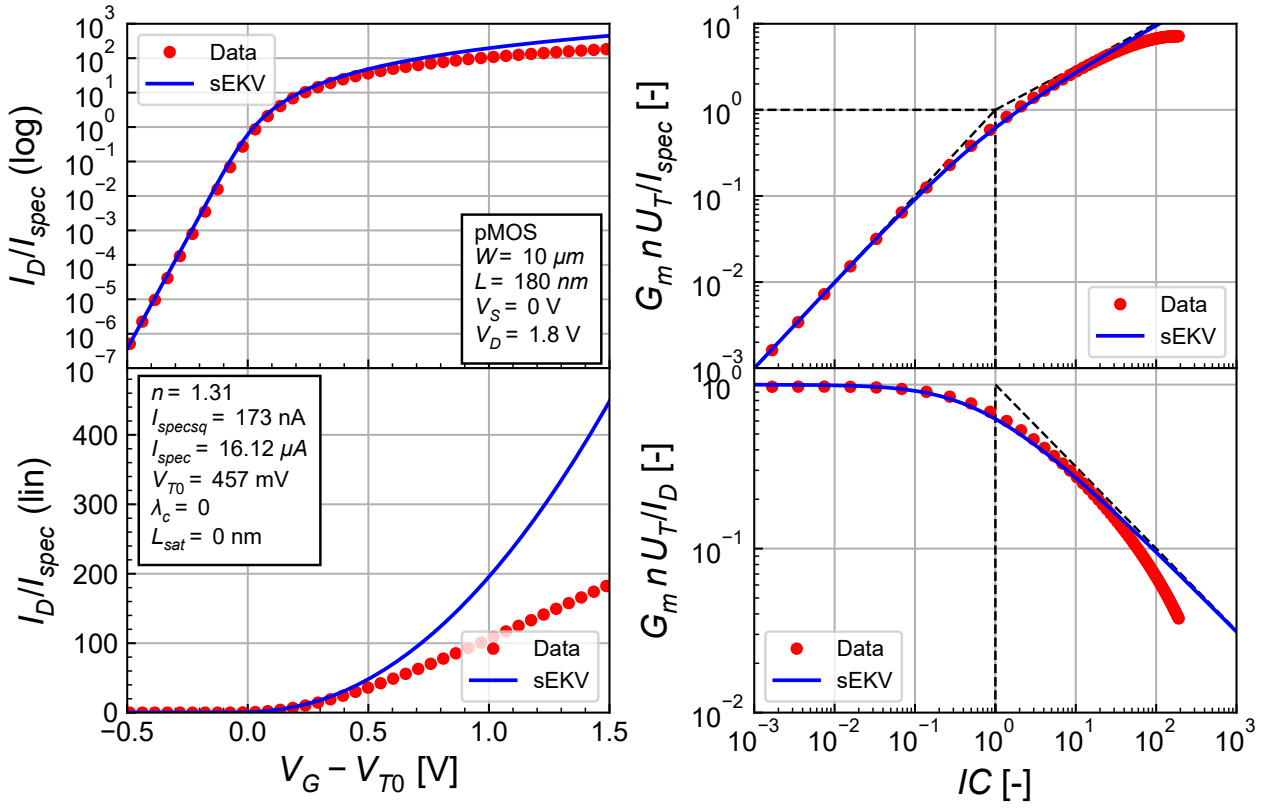


Figure 3.41: Result for the parameters calculated from the EKV 2.6 CM parameters with $\lambda_c = 0$.

The fit is good in weak inversion but not that good in moderate inversion and completel off in strong inversion. We can improve it by including $L_{sat} = 0\ nm$ resulting in Figure 3.42.

The fit is now very good. The resulting parameters are summarized in Table 3.4.

Table 3.4: Extraction of the sEKV parameters from the EKV 2.6 parameters.

Type	n	$I_{spec\square}\ [nA]$	$V_{T0}\ [mV]$	λ_c	$L_{sat}\ [nm]$	Comment
pMOS	1.36	250	491	0.000	0.000	direct with $\lambda_c = 0$
pMOS	1.36	200	490	0.139	15.000	direct with $\lambda_c > 0$
pMOS	1.33	200	484	0.139	15.000	curve fitting with $\lambda_c > 0$
pMOS	1.31	173	457	0.111	12.000	Calculated from EKV 2.6

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Table 3.4: Extraction of the sEKV parameters from the EKV 2.6 parameters.

Type	n	$I_{spec\square}$ $[nA]$	V_{T0} $[mV]$	λ_c	L_{sat} $[nm]$	Comment

3.7 Check Parameters on Long-channel Transistor

It is interesting to check how the parameters extracted for the short-channel transistor work on the long-channel transistor. For this we use the paramters extracted from the direct extraction with $\lambda_c > 0$ (second line in Table 3.4). The result is presented in Figure 3.43.

The fit is reasonable in all the regions.

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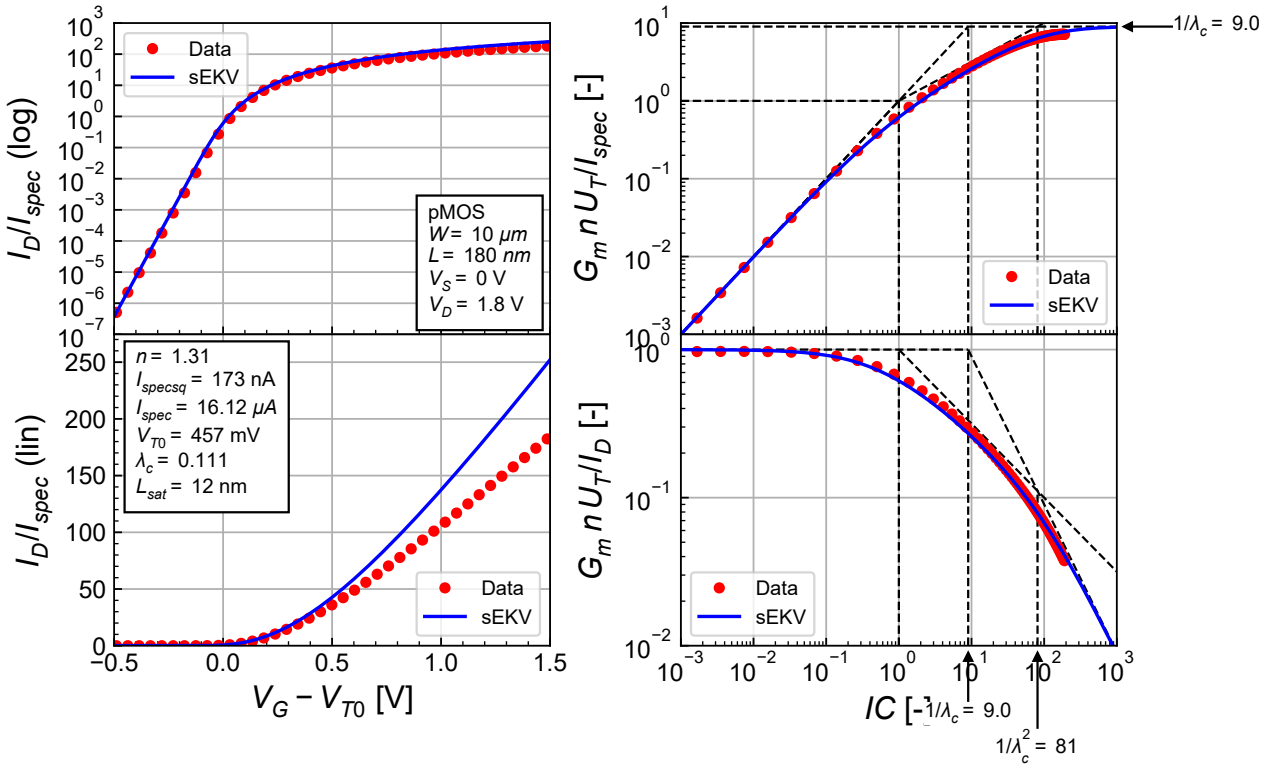


Figure 3.42: Result for the parameters calculated from the EKV 2.6 CM parameters.

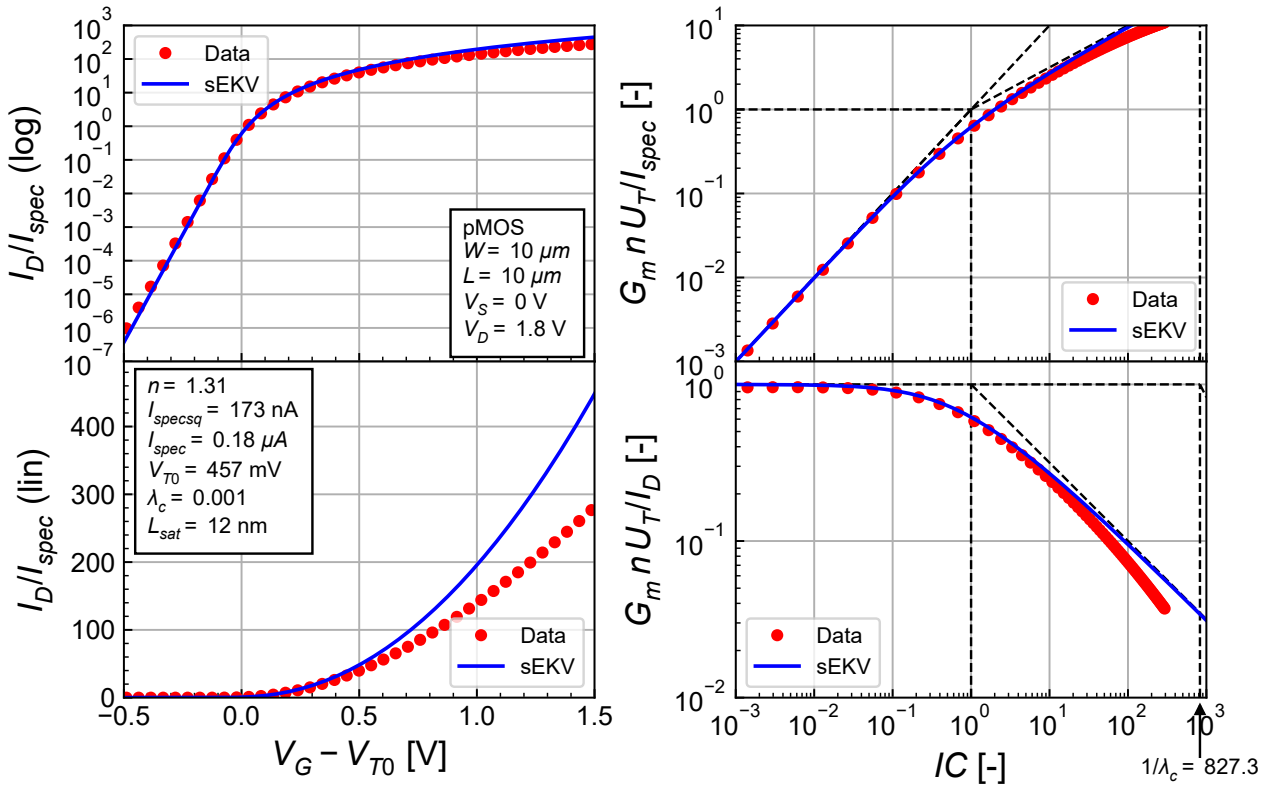


Figure 3.43: Check the parameters extracted for the short-channel transistor on a long-channel transistor.

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4 Output characteristic

4.1 Generating and importing the data

The data used for the sEKV parameters extraction is generated by simulation using the EKV 2.6 compact model for a generic 180nm process. We present the I_D-V_G and G_m-V_G data below.

4.1.1 I_D and G_{ds} versus V_D

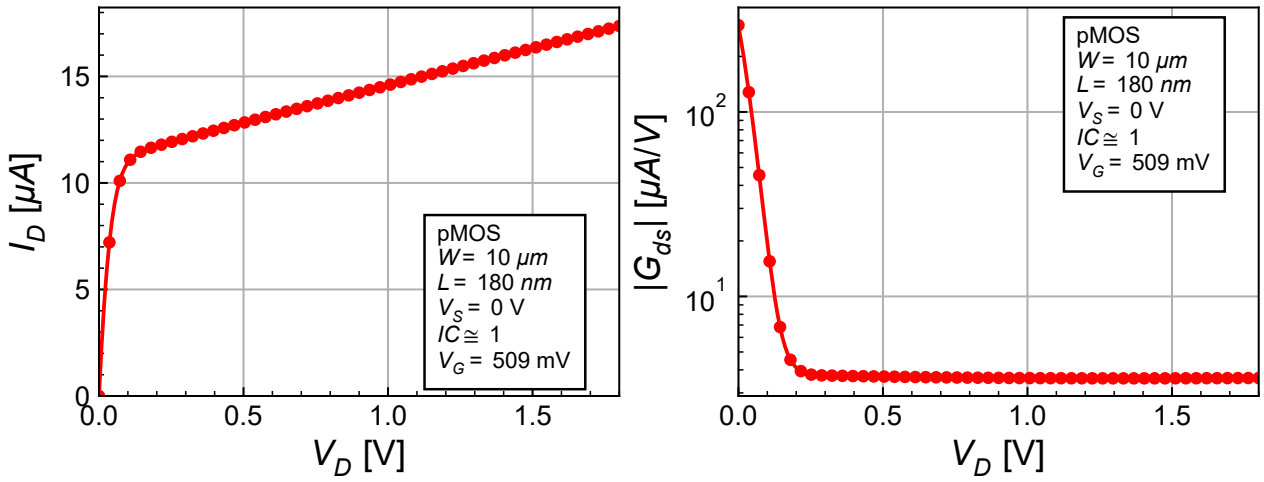


Figure 4.1: Imported I_D-V_D and G_{ds}-V_D.

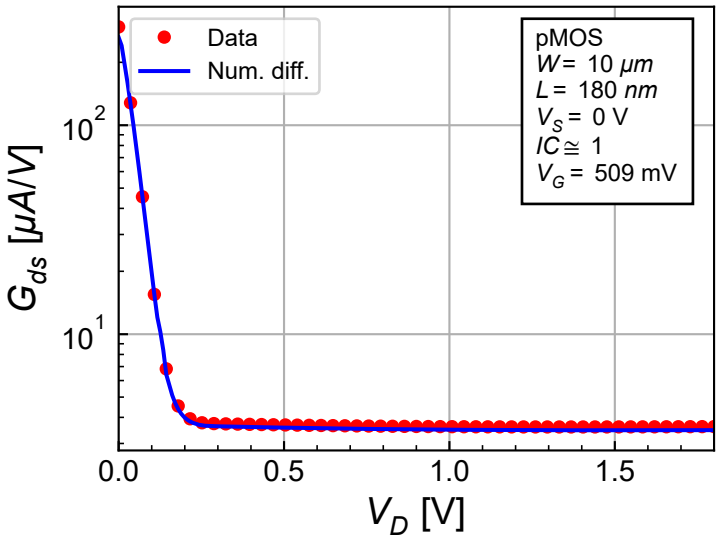


Figure 4.2: Check imported G_{ds} against computed G_{ds}.

The output conductance calculated by differentiating the large-signal I_D-V_D matches the value extracted from the EKV 2.6 compact model. We will keep the value from EKV 2.6.

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4.2 Filtering the outliers

In order to extract the CLM parameter in saturation, we will now filter out the points that correspond to the linear region. We can do this easily by looking at the G_{ds} versus V_D characteristic as shown in Figure 4.3.

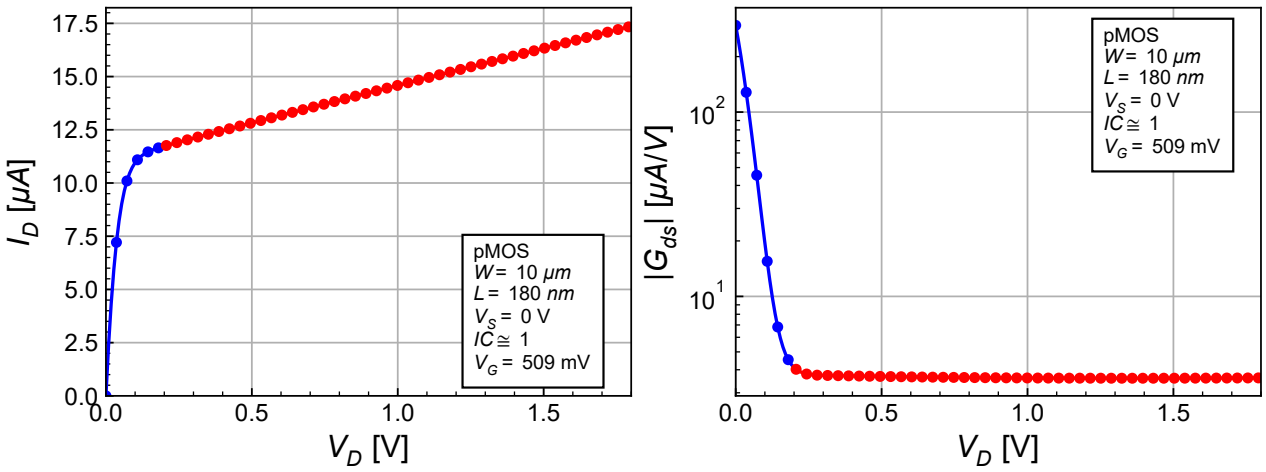


Figure 4.3: Filtering the outliers from I_D-V_G and G_{ds}-V_D.

4.3 Extracting the CLM parameter

The simple channel length modulation (CLM) of the output conductance in saturation is approximated by

$$G_{ds} = \frac{I_{D,sat}}{V_E} \tag{4.1}$$

where $I_{D,sat}$ is the drain current in saturation and

$$V_E = \lambda \cdot L_{eff} \tag{4.2}$$

is the Early voltage or CLM voltage which is proportionnal to the effectve length. This corresponds to a linear approximation of the drain current in saturation given by

$$I_{D,sat} \cong G_{ds} \cdot (V_D - V_E) \tag{4.3}$$

The value I_{D0} corresponds to the intercept

$$I_{D0} = -G_{ds} \cdot V_E \tag{4.4}$$

which is positive since V_E is negative.

The linear regression approximation of the drain current in saturation is shown in Figure 4.4 by the dashed black line. We get a rather high output conductance $G_{ds} = 3.506\ \mu A/V$ and hence low value of the λ parameter with $\lambda = 17.543\ V/\mu m$ which is expected for the short transistor. This translates into a poor intrinsic voltage gain G_m/G_{ds}. The output conductance parameters are summarized in Table 4.1.

As shown in the right plot of Figure 4.4, the output conductance is approximated by a constant value represented by the dashed black line. It is a poor approximation of the output conductance but good enough for starting a design. Note that, in analog circuit design, having a good fit of the output conductance is less important than having a good fit of the transconductance because often we just need the intrinsic voltage gain to be large enough for the circuit to perform correctly, while we

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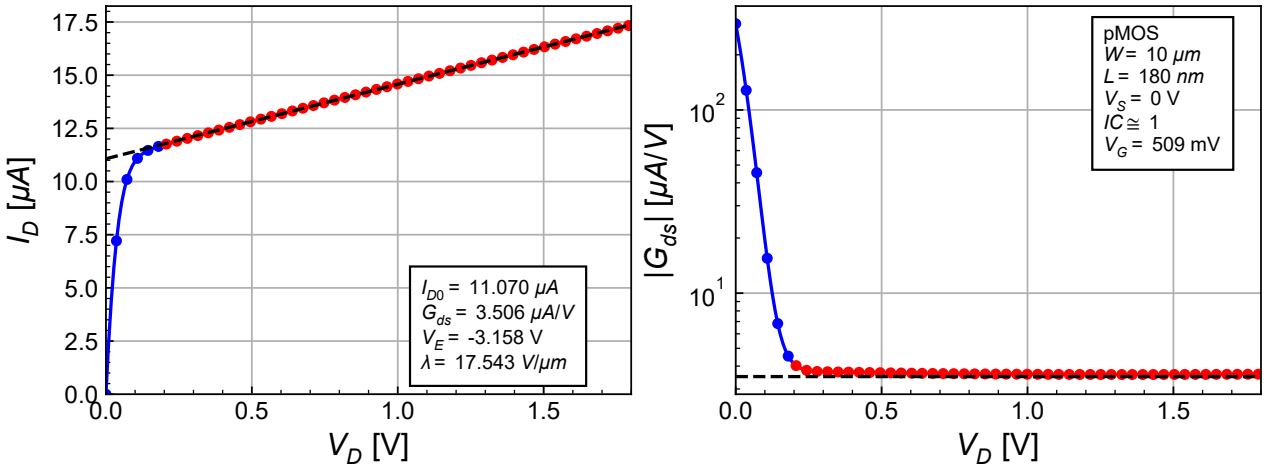


Figure 4.4: Extraction of the CLM parameter λ.

want to have a good estimation of the parameters depending on the transconductance, such as the gain-bandwidth product or the thermal noise.

Table 4.1: CLM parameters extracted in moderate inversion.

Type	L_{eff} [μm]	IC	G_{ds} [$\mu A/V$]	I_{D0} [μA]	V_E [V]	λ [$V/\mu m$]	Comment
nMOS	0.108	1	3.506	11.070	-3.158	17.543	extracted in MI

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5 Noise

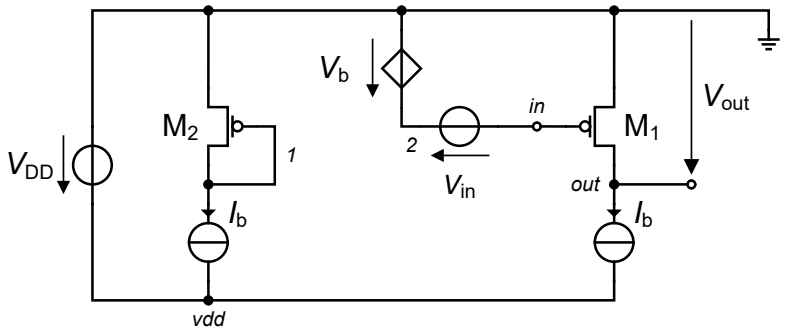


Figure 5.1: Schematic used for the noise simulation.

We now will check the noise model including the thermal and flicker noise. The simulations will be performed with the help of the schematic shown in Figure 5.1. Transistor M_1 is the active transistor and M_2, which is identical to M_1, is used to set the gate voltage of M_1 such that its current is equal to I_b.

5.1 Setting the bias conditions

Table 5.1: Transistor size and bias information.

Transistor	$W\ [\mu m]$	$L\ [nm]$	$I_D\ [\mu A]$	$I_{spec}\ [\mu A]$	IC	$V_G - V_{T0}\ [mV]$	$V_{DSsat}\ [mV]$
M1	10.00	180	16.117	16.117	1.0	15	116
M2	10.00	180	16.117	16.117	1.0	15	116

We choose an inversion coefficient $IC = 1$, which for the selected transistor size leads to a bias current $I_b = 16.117\ \mu A$.

Table 5.2: Transistor small-signal and thermal noise parameters.

Transistor	$G_{spec}\ [\mu A/V]$	$G_{ms}\ [\mu A/V]$	$G_m\ [\mu A/V]$	$G_{ds}\ [\mu A/V]$	γ_n
M1	622.895	384.971	294.763	8.507	0.736
M2	622.895	384.971	294.763	8.507	0.736

5.2 Extract operating point information

The node voltages are given in Table 5.3.

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Table 5.3: Node voltages.

Node	Voltage
vdd	-1.8
in	-0.512282
out	-0.512282
1	-0.512282
2	-0.512282
1	-0.512282

Table 5.4: Operating point information extracted from ngspice .op file for each transistor.

Transistor	I_D $[\mu A]$	I_{spec} $[\mu A]$	IC	n	V_{Dsat} $[mV]$
M1	16.117	16.933	0.970	1.31	154
M2	16.117	16.933	0.970	1.31	154

Table 5.5: Small-signal operating point information extracted from ngspice .op file for each transistor.

Transistor	G_m $[\mu A/V]$	G_{ms} $[\mu A/V]$	G_{ds} $[\mu A/V]$
M1	295.081	391.025	4.526
M2	295.081	391.025	4.526

Table 5.6: Noise operating point information extracted from ngspice .op file for each transistor.

Transistor	R_n $[k\Omega]$	$\sqrt{S_{ID,th}}$ $[nA/\sqrt{Hz}$	γ_n [-]	$\sqrt{S_{ID,fl}}$ at 1Hz $[nA/\sqrt{Hz}]$
M1	2.587	6.548	0.763	85720.6
M2	2.587	6.548	0.763	85720.6

5.3 Input-referred noise PSD

To compare the simulated noise to the theoretical noise PSD, we can simulate the circuit shown in Figure 5.1. The output- and input-referred noise PSD will include both contributions of M_1 and M_2. Since M_1 and M_2 are identical and have the same bias current, we can simply divide the output- and input-referred noise PSD by a factor 2. In ngspice to get the noise PSD we need to set `setplot noise1`. We can then export the noise data using `wrdata $inputdir/noise.nz.dat inoise_spectrum onoise_spectrum`. The simulated input-referred noise PSD is compared to the theoretical one in Figure 5.2.

We see a perfect match between the theroretical and simulated total input-referred noise PSD.

We can extract the individual thermal and flicker noise contributions of M_1 only by the following instructions `save onoise_n1 onoise_n1_flicker onoise_n1_thermal` and saving the raw data with `write $inputdir/noise.nz.raw`. The resulting PSD are plotted in Figure 5.3.

We see that the simulation matches the theoretical estimation, particularly for the flicker noise. The simulated thermal noise is slightly larger than the theoretical estimation. This comes from the slightly smaller simulated transconductance compared to the theoretical value.

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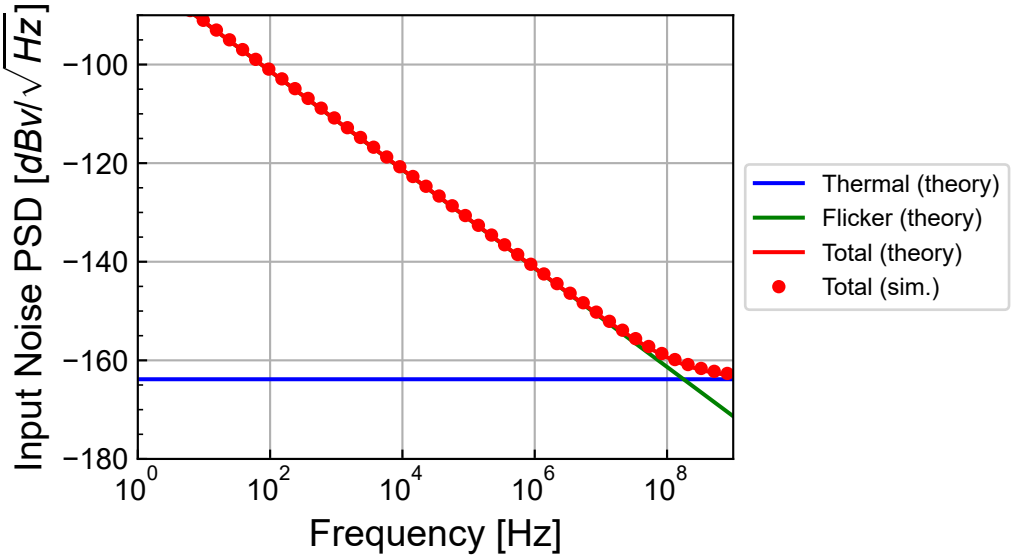


Figure 5.2: Simulated input-referred noise PSD of M_1 compared to theoretical estimation from `inoise_spectrum`.

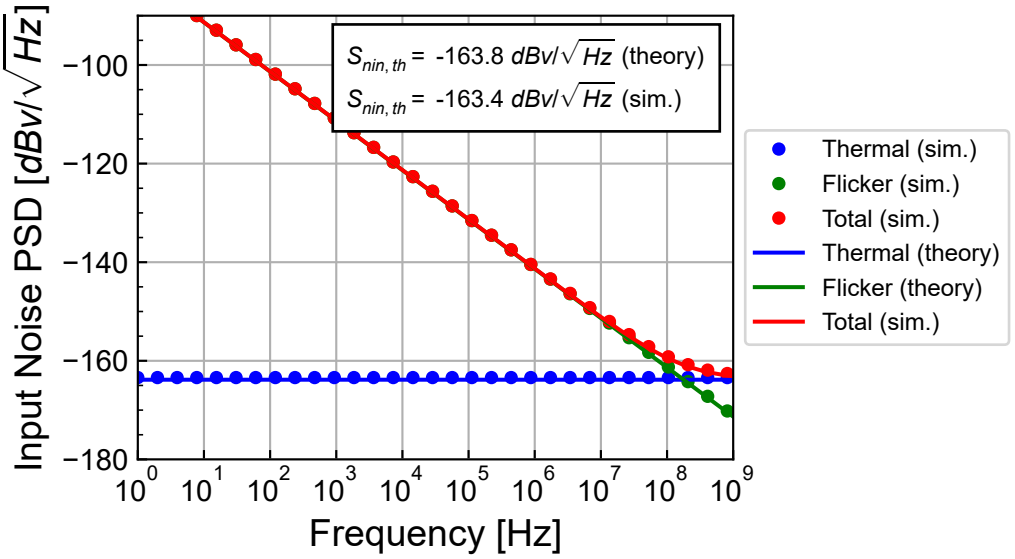


Figure 5.3: Simulated input-referred noise PSD of M_1 compared to theoretical estimation.

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6 Conclusion

This notebook presented different approaches to extract the sEKV parameters for a short-channel pMOS transistor from a generic 180 nm bulk CMOS process. The data was first obtained by simulations using the EKV 2.6 compact model.

The sEKV parameters were first extracted using a direct extraction methodology with $\lambda_c = 0$. This results in a good fit over all regions of operation except in very strong inversion because of the combined effects of velocity saturation and mobility reduction due to the vertical field.

The direct extraction approach is then used to also extract the velocity saturation parameter λ_c. After some fine tuning we can get a very good fit in all regions of operation including strong inversion.

The sEKV parameters are then extracted using curve fitting which immediately gives a good result in all regions of operation with values that are close to those extracted with the direct extraction methodology.

The parameters for the output conductance due to CLM are then extracted in moderate inversion.

Finally, we also have checked that the thermal and flicker noise match the theoretical model.

Overall we have shown that sEKV can fit the large- and small-signal data very well and that the noise is consistent with EKV 2.6 compact model for this short-channel nMOS transistor.

> **!** Important
>
> Although the extraction of the sEKV parameters using either direct extraction or by optimization give very good fit, we will use the parameters that are calculated from the EKV 2.6 compact model parameters adding L_{sat} and the CLM parameter λ. Indeed, this gives simulation results that are more consistent with the theoretical prediction, particularly for the inversion coefficient and the transconductances.

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